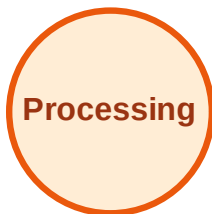


BFS Traversal

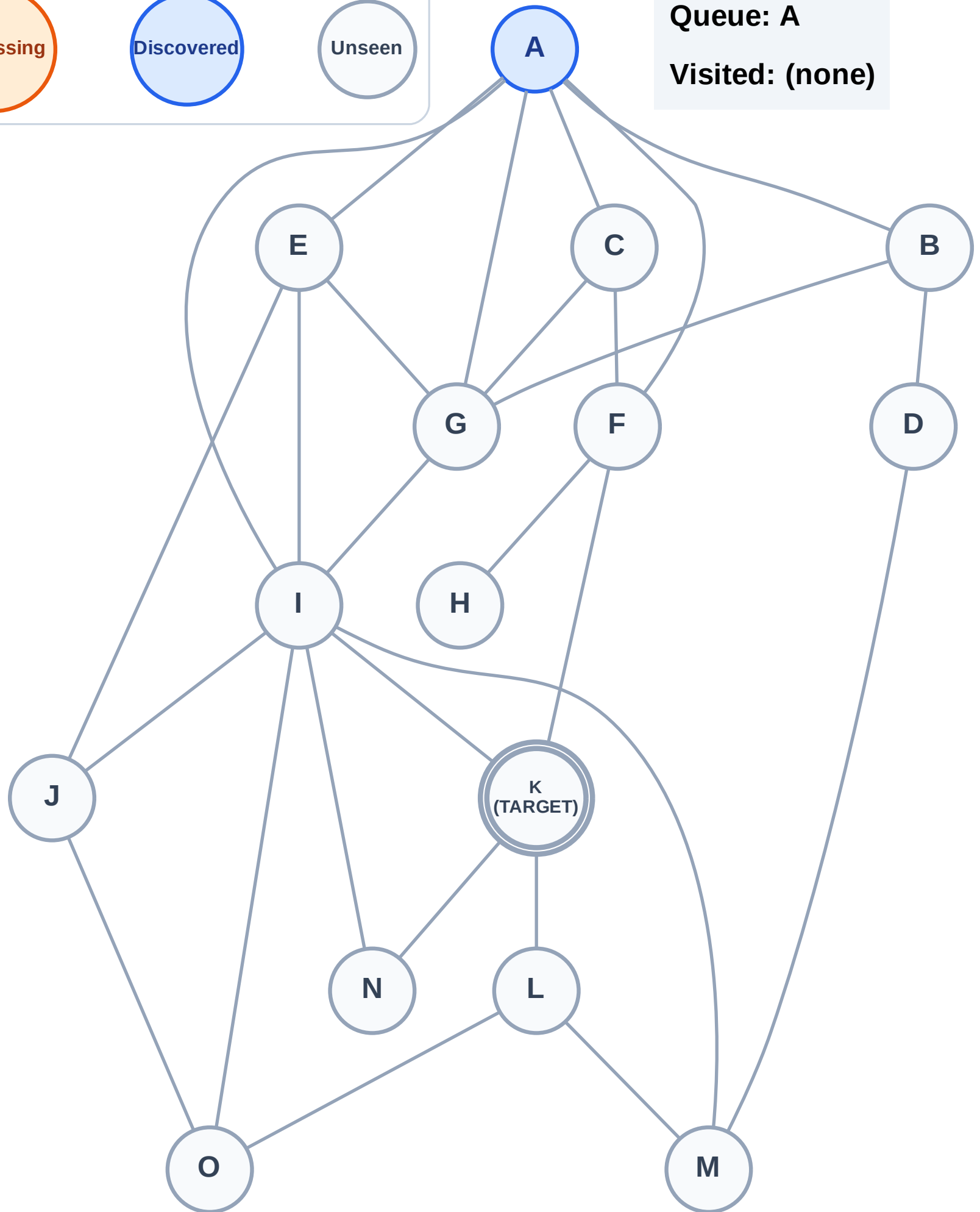
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

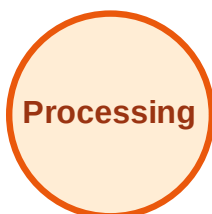
Visited: (none)



BFS Traversal

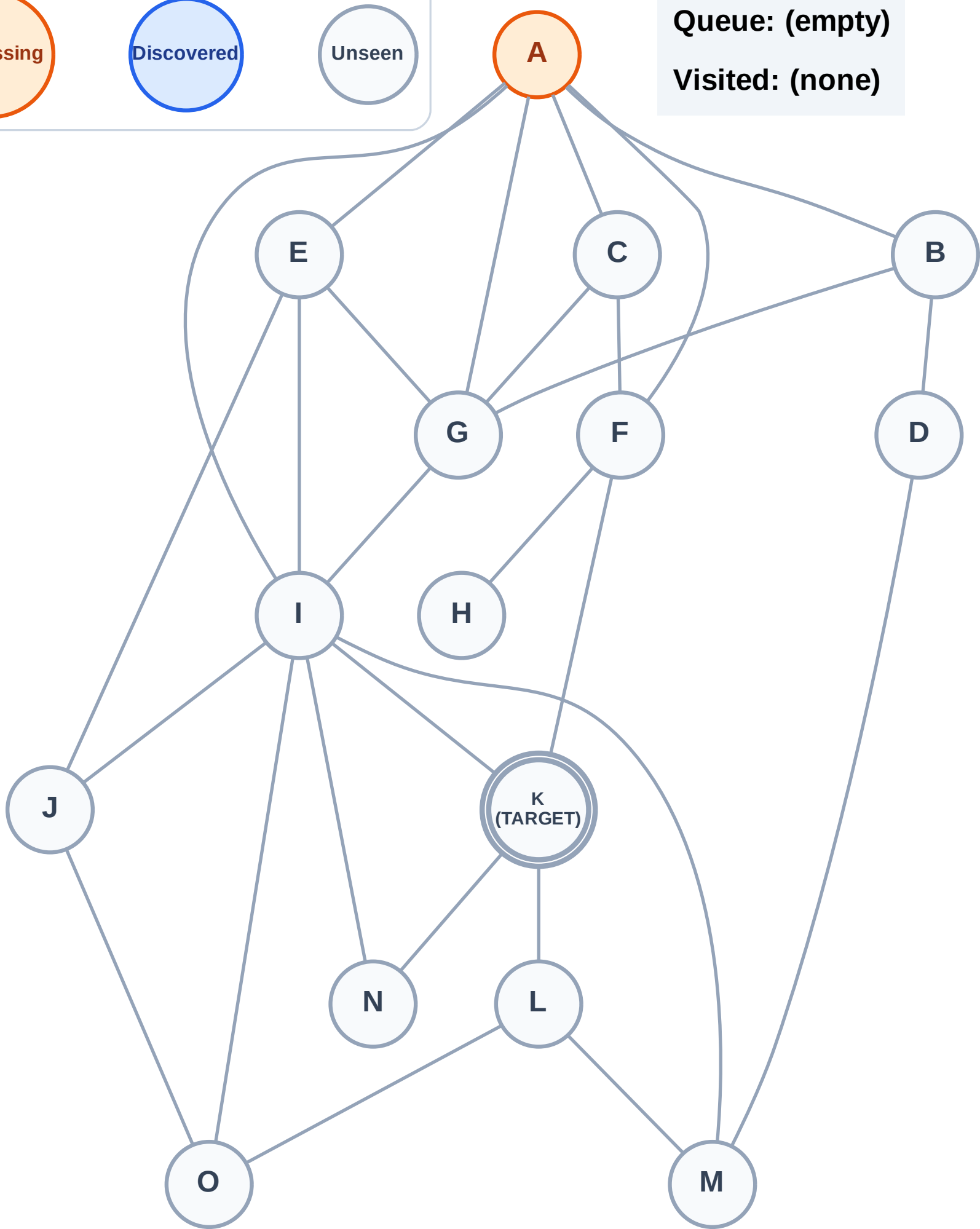
Step 2: Process node A

Legend



Queue: (empty)

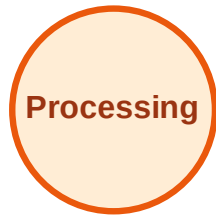
Visited: (none)



BFS Traversal

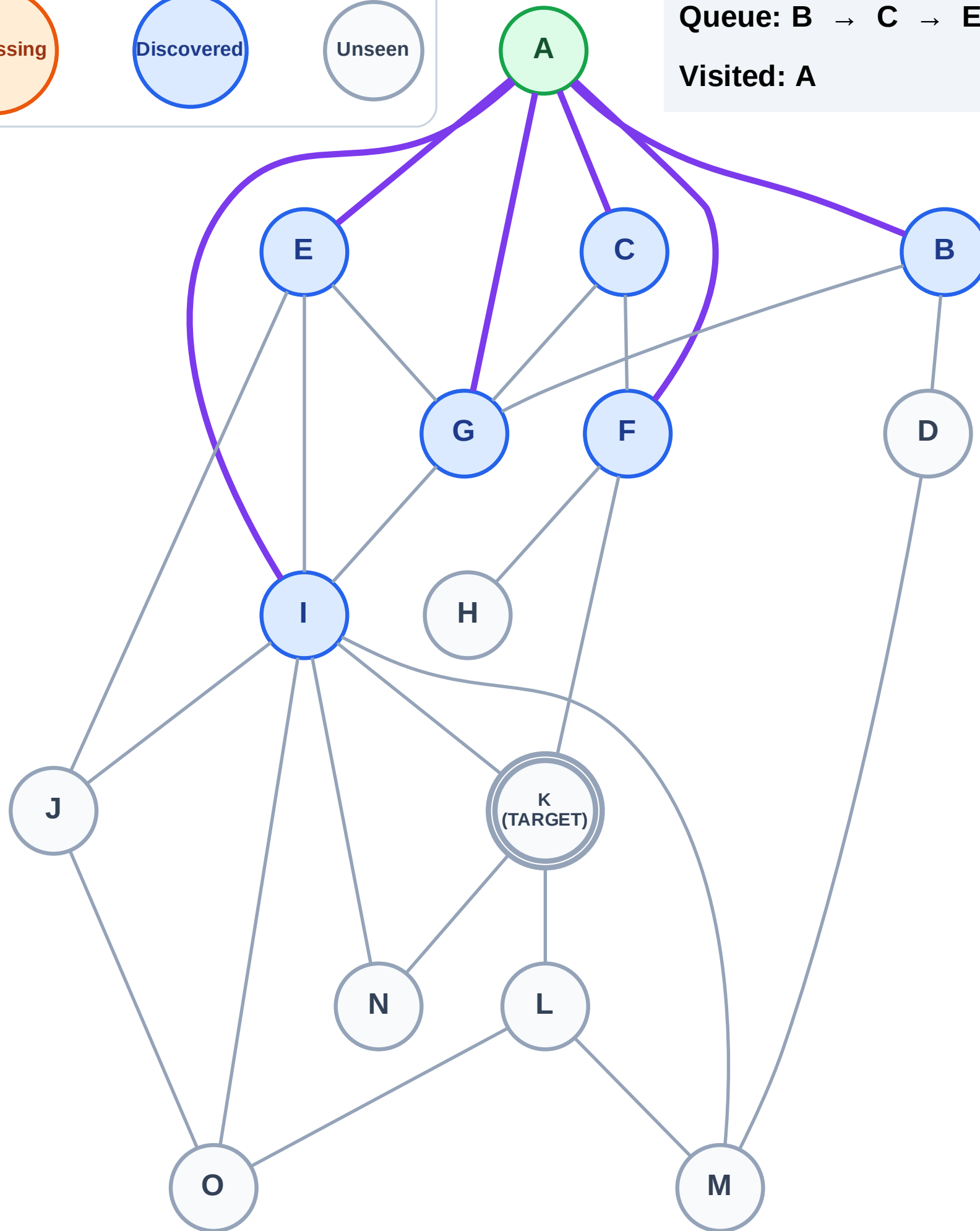
Step 3: Discover B, C, E, F, G, I from A

Legend



Queue: B → C → E → F → G → I

Visited: A



BFS Traversal

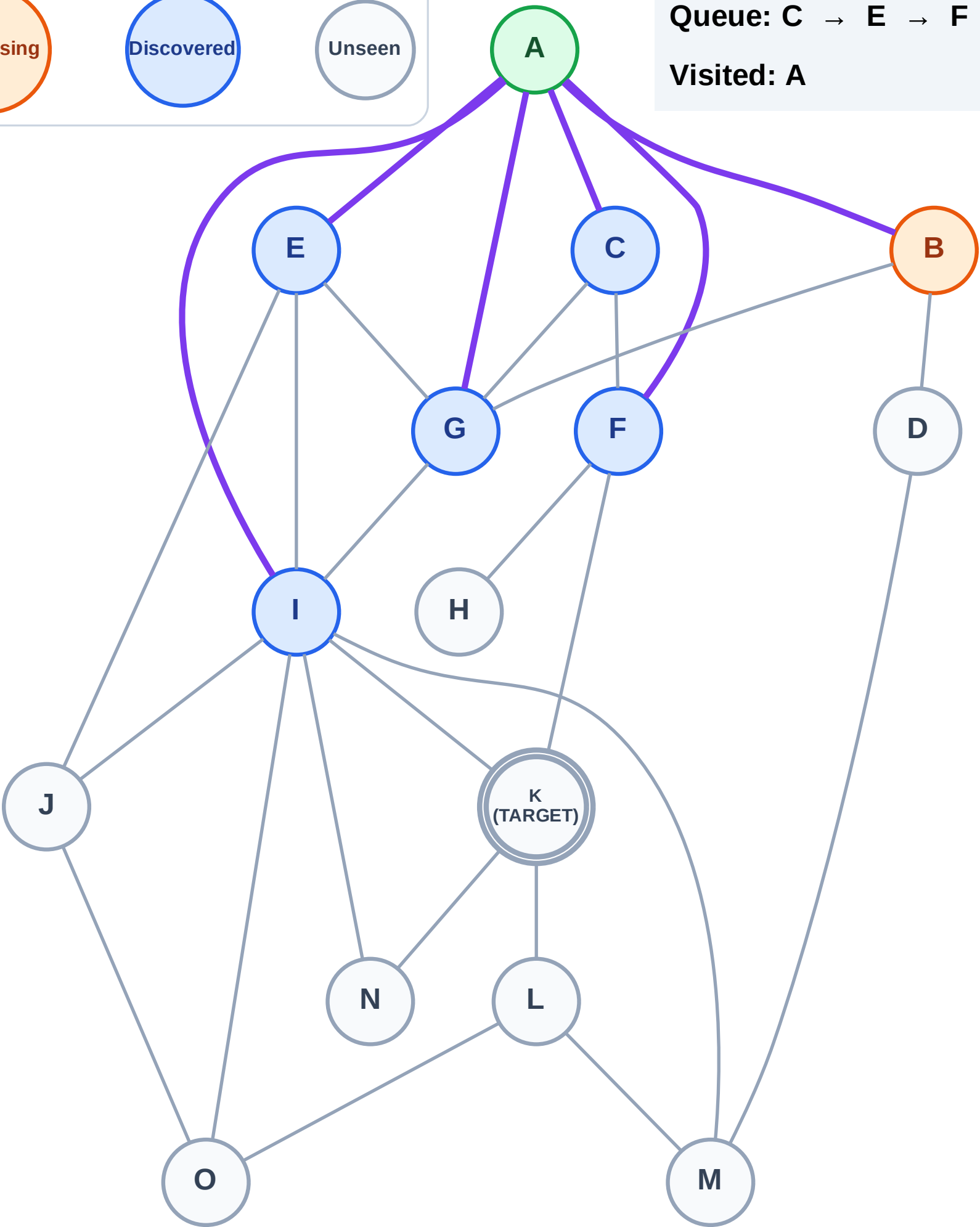
Step 4: Process node B

Legend



Queue: C → E → F → G → I

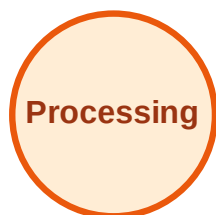
Visited: A



BFS Traversal

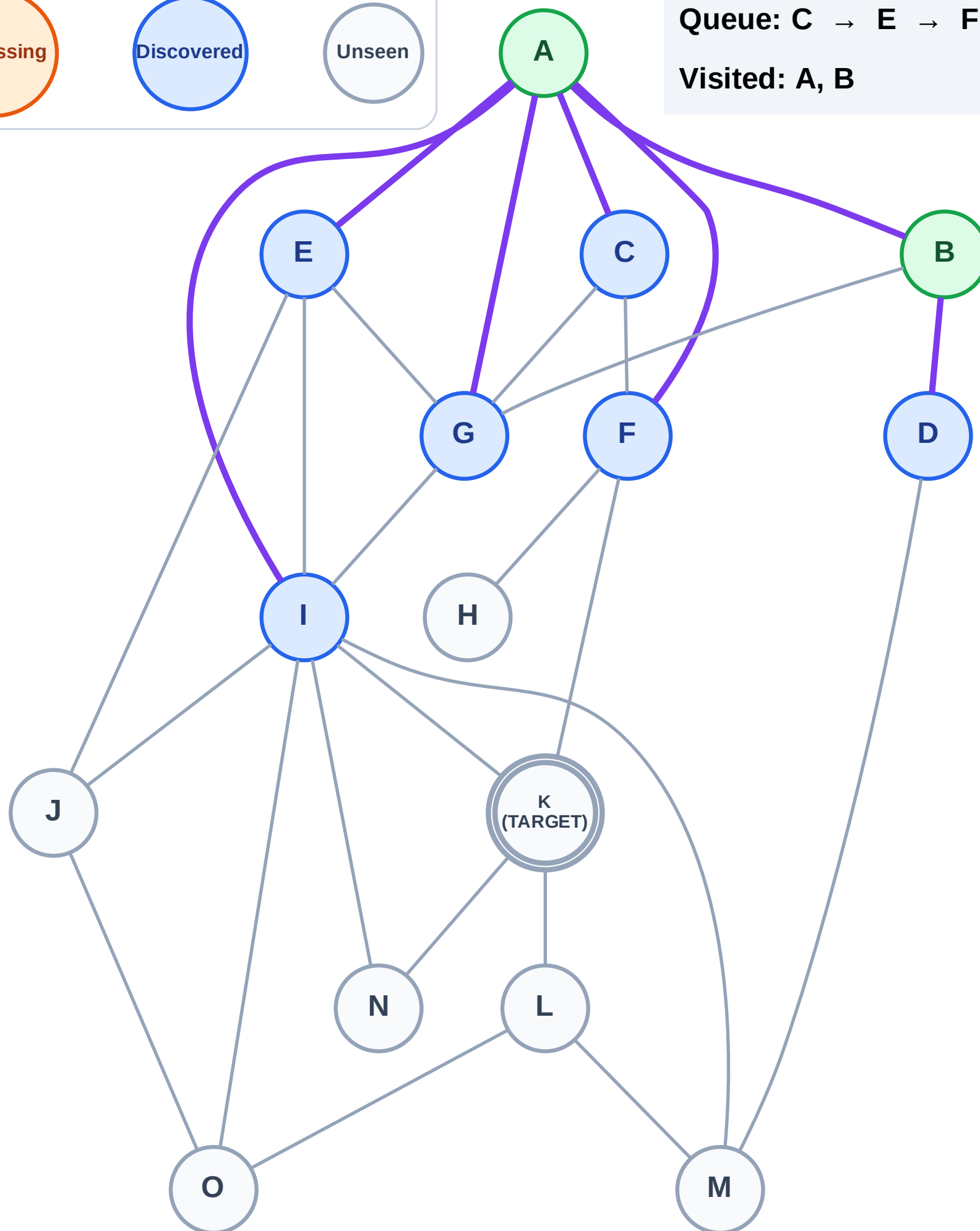
Step 5: Discover D from B

Legend



Queue: C → E → F → G → I → D

Visited: A, B



BFS Traversal

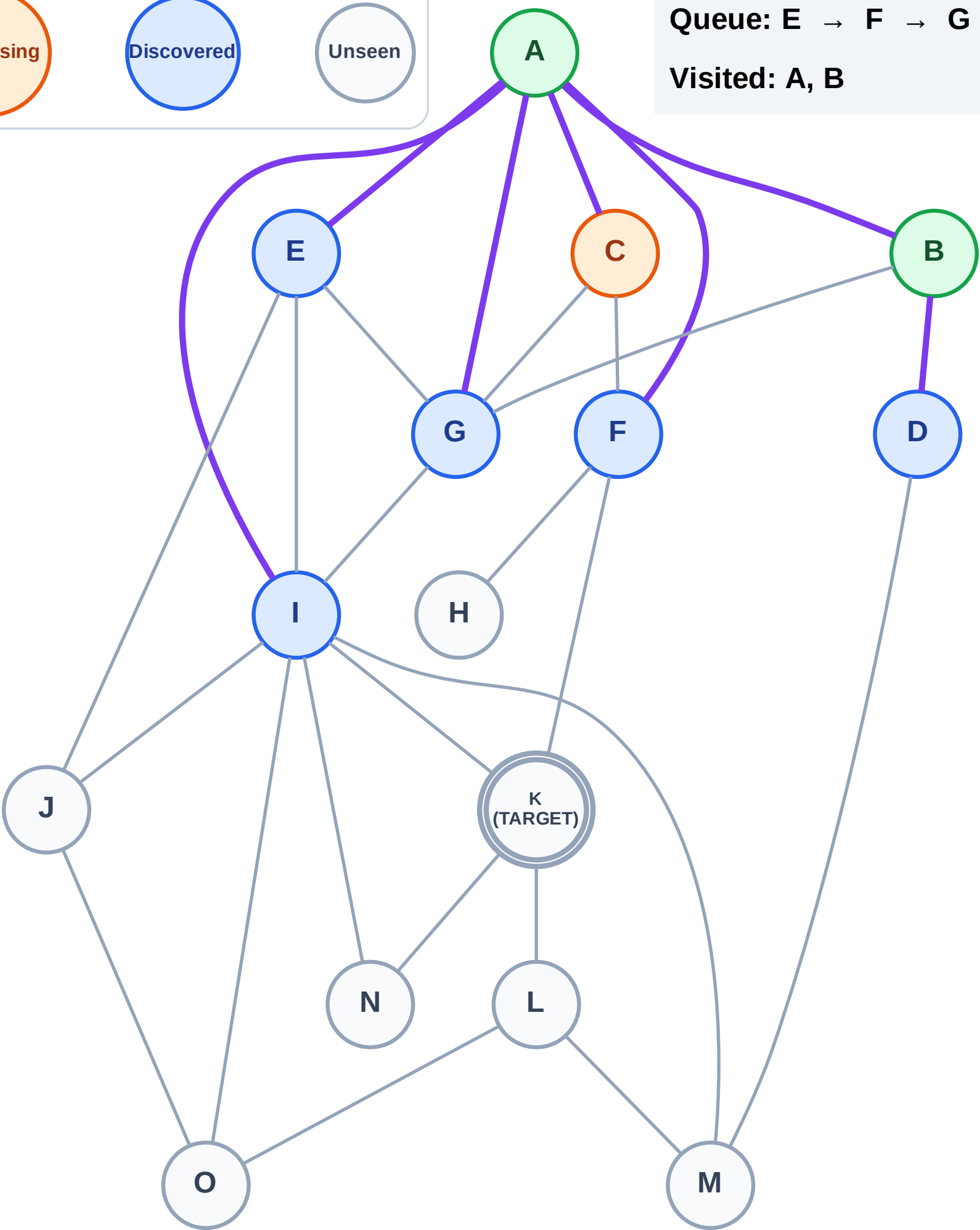
Step 6: Process node C

Legend



Queue: E → F → G → I → D

Visited: A, B



BFS Traversal

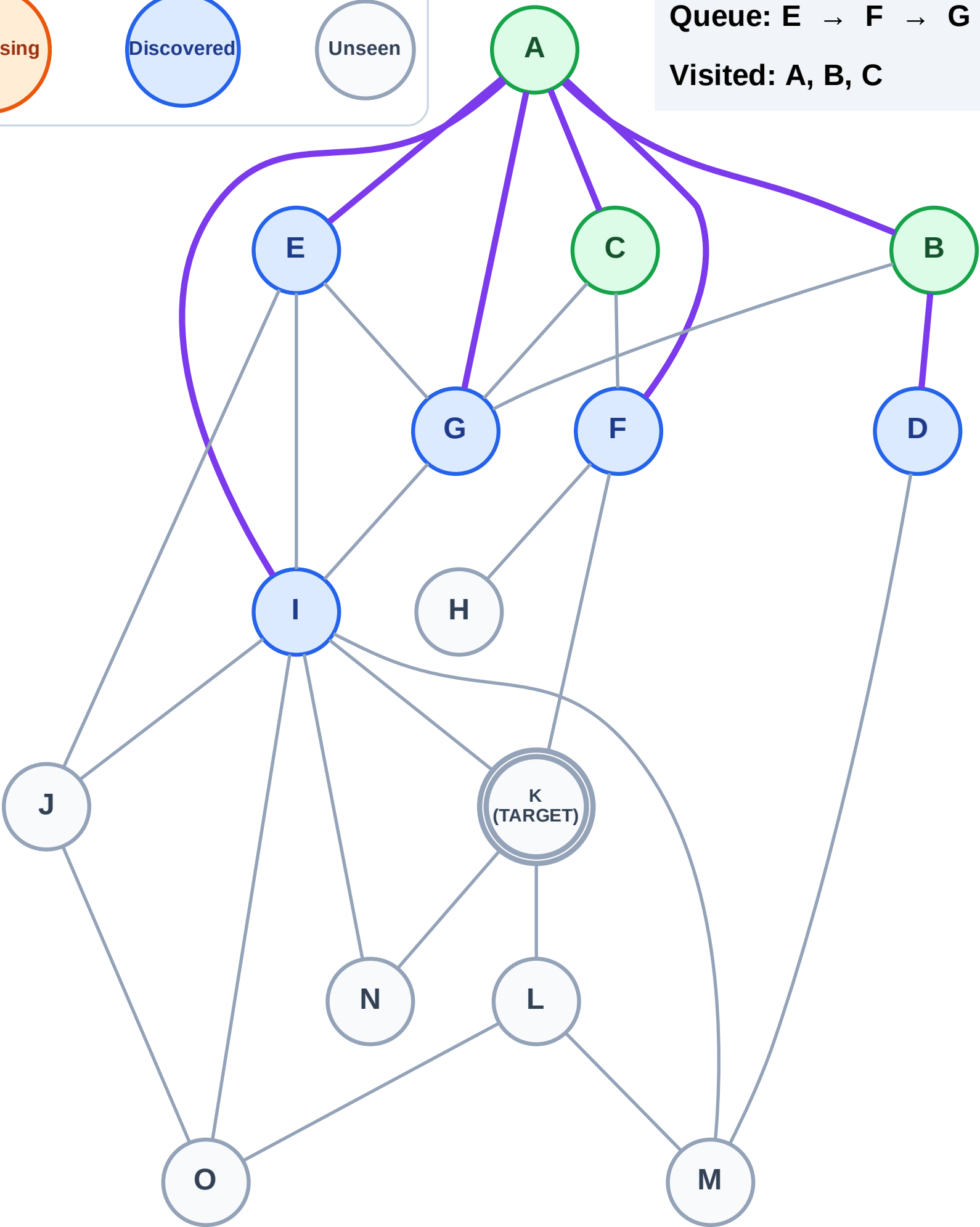
Step 7: No new neighbors from C

Legend



Queue: E → F → G → I → D

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

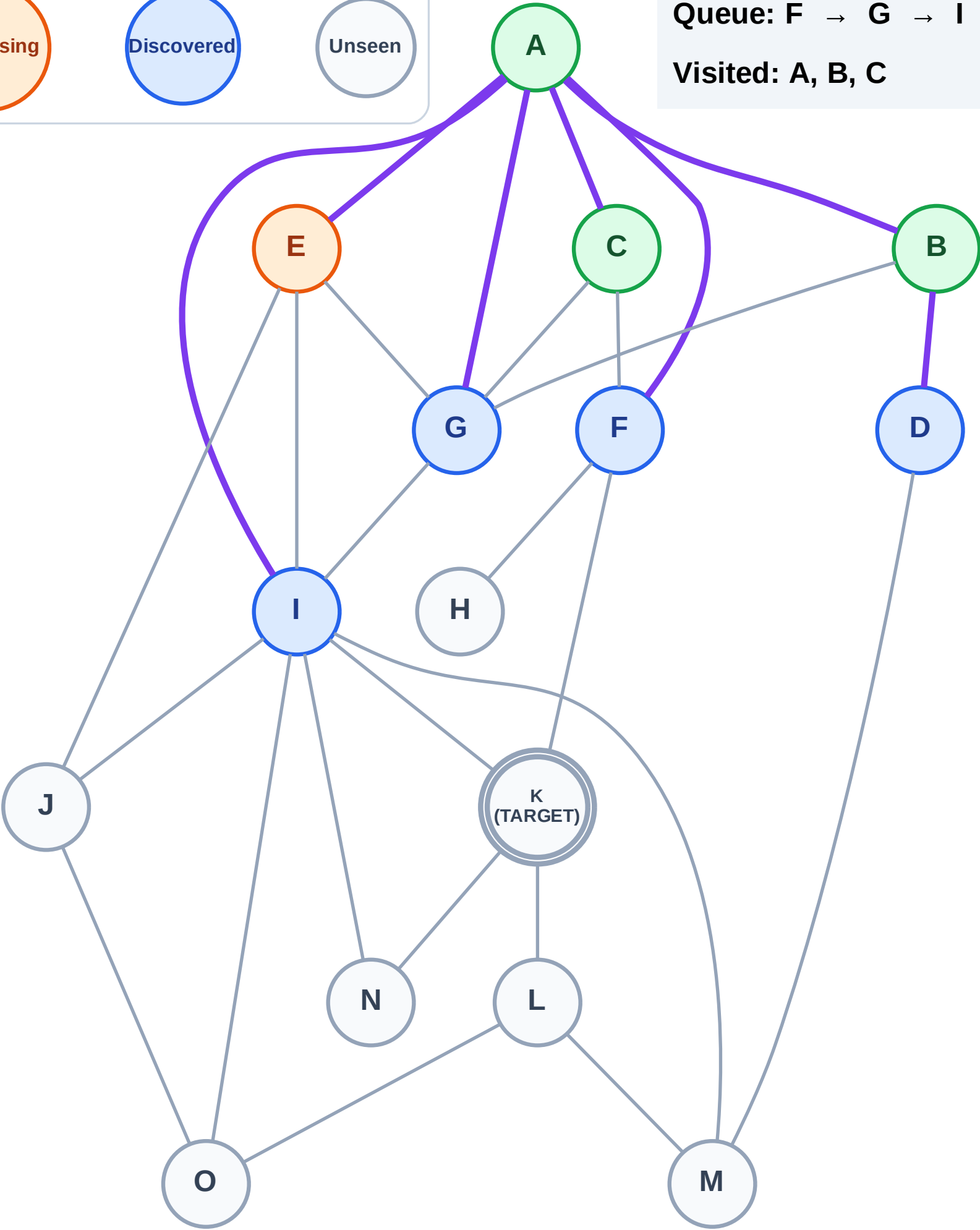
Processing

Discovered

Unseen

Queue: F → G → I → D

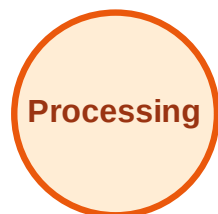
Visited: A, B, C



BFS Traversal

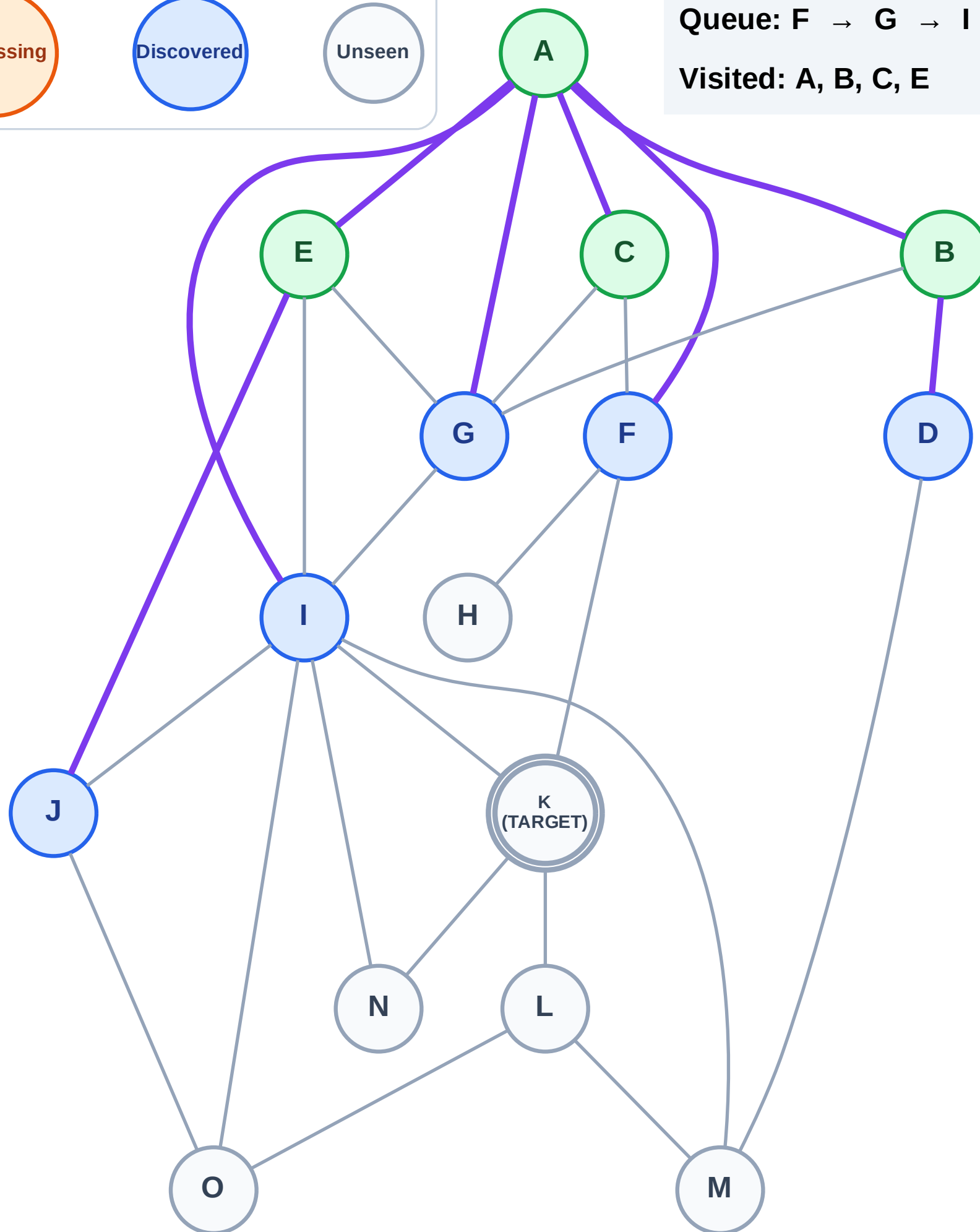
Step 9: Discover J from E

Legend



Queue: F → G → I → D → J

Visited: A, B, C, E



BFS Traversal

Step 10: Process node F

Legend

Complete

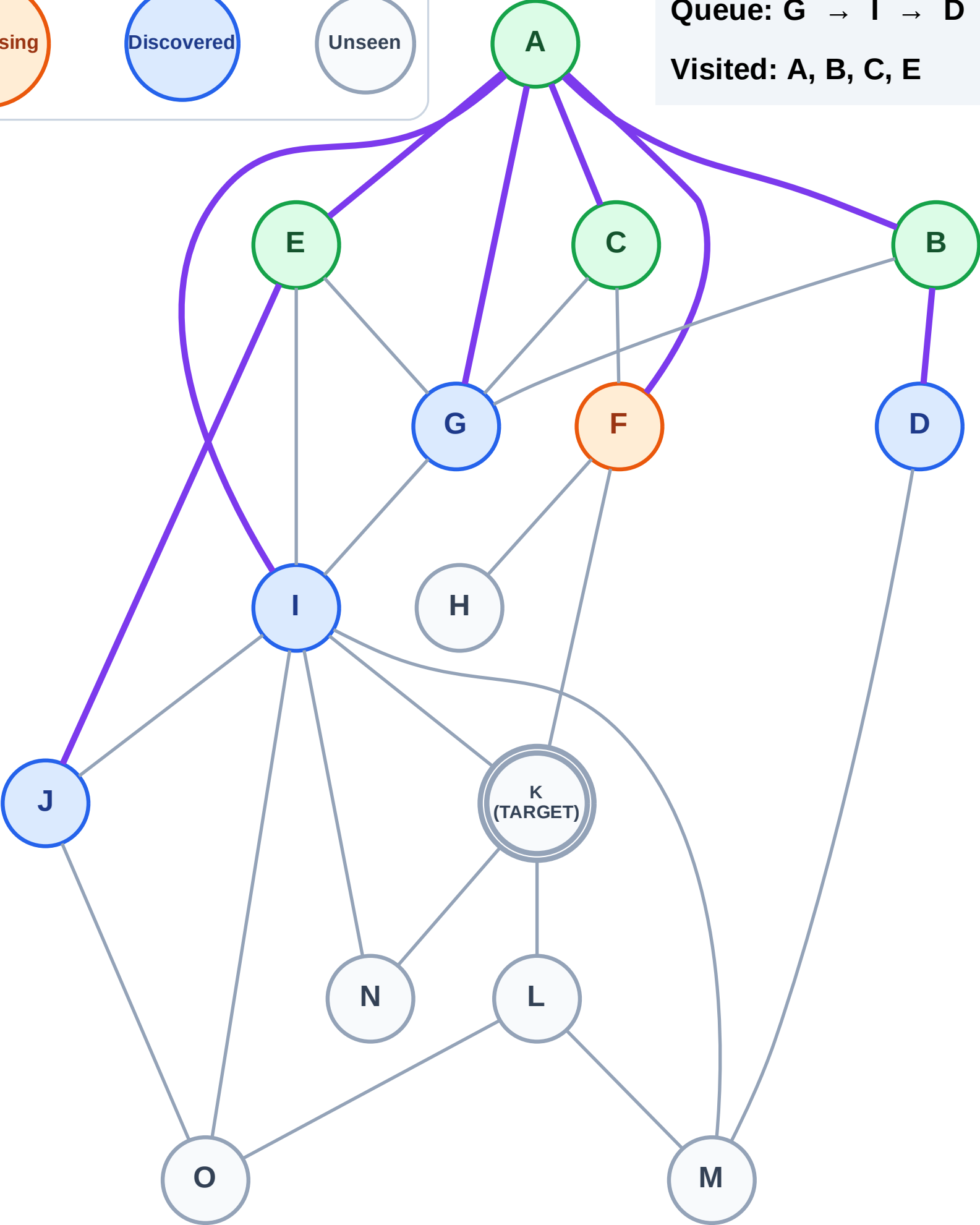
Processing

Discovered

Unseen

Queue: G → I → D → J

Visited: A, B, C, E



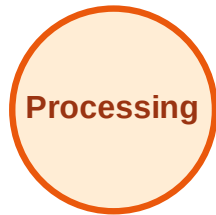
BFS Traversal

Step 11: Discover H, K from F

Legend



Complete



Processing



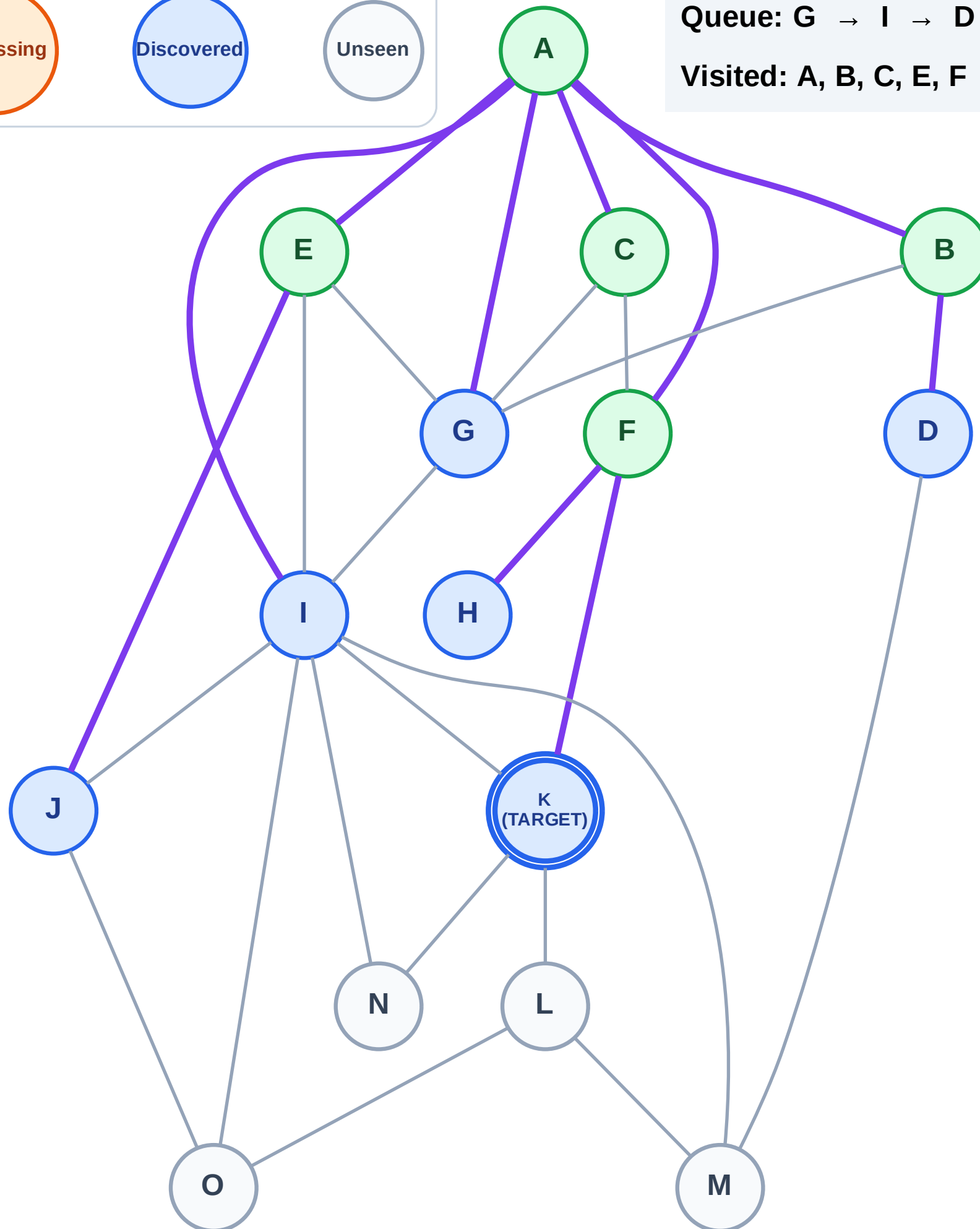
Discovered



Unseen

Queue: G → I → D → J → H → K

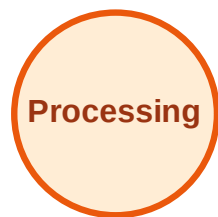
Visited: A, B, C, E, F



BFS Traversal

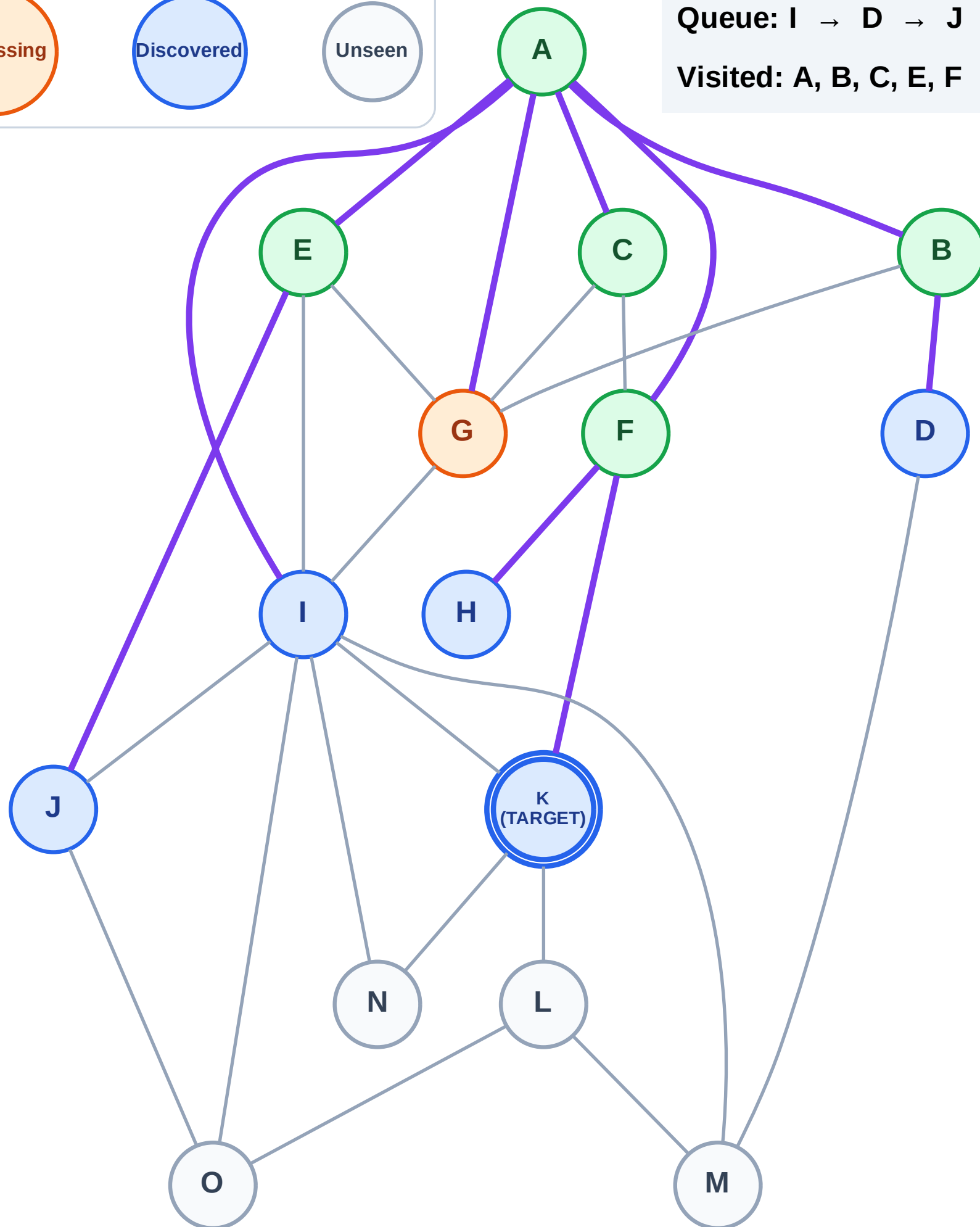
Step 12: Process node G

Legend



Queue: I → D → J → H → K

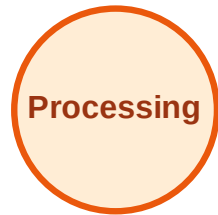
Visited: A, B, C, E, F



BFS Traversal

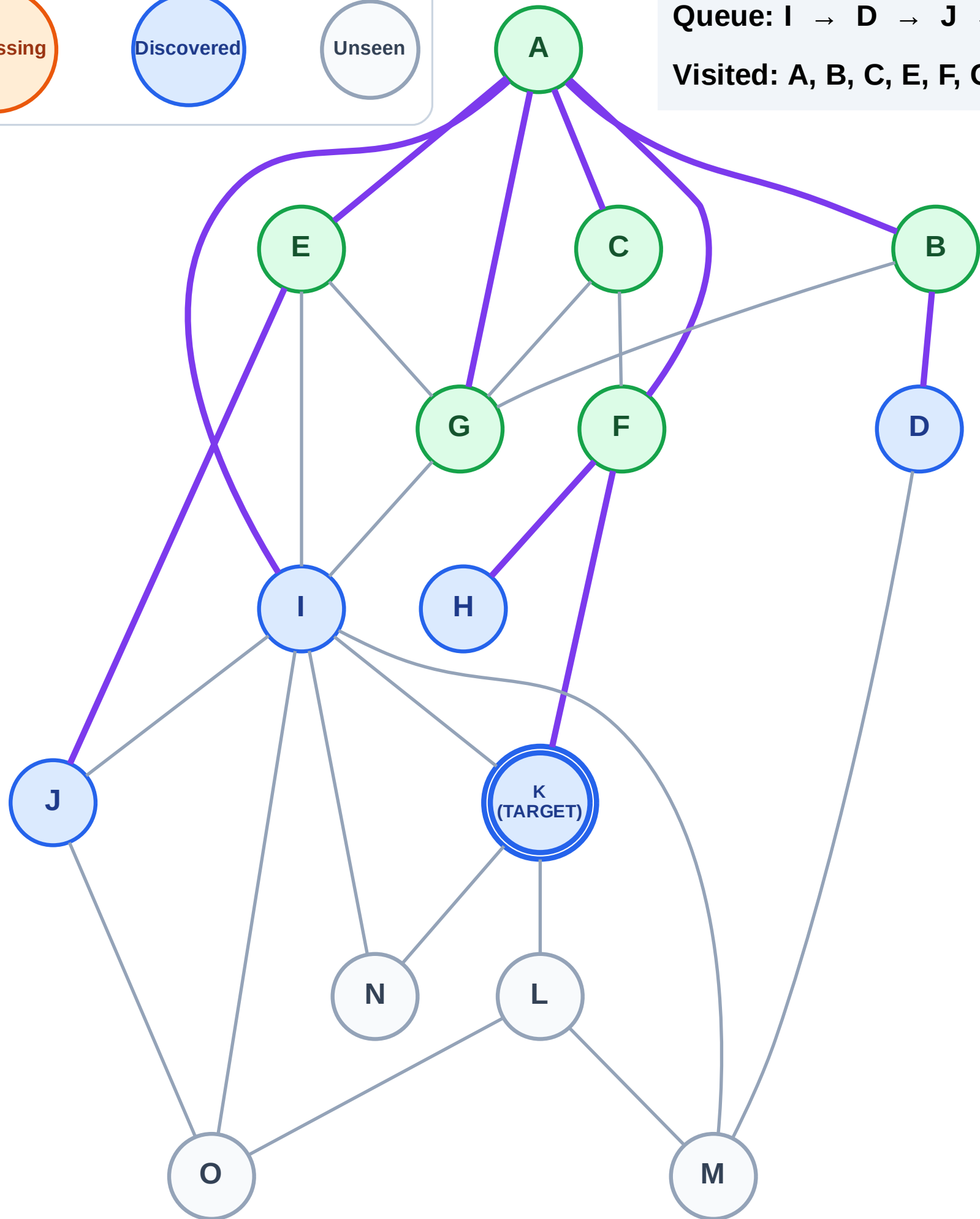
Step 13: No new neighbors from G

Legend



Queue: I → D → J → H → K

Visited: A, B, C, E, F, G



BFS Traversal

Step 14: Process node I

Legend

Complete

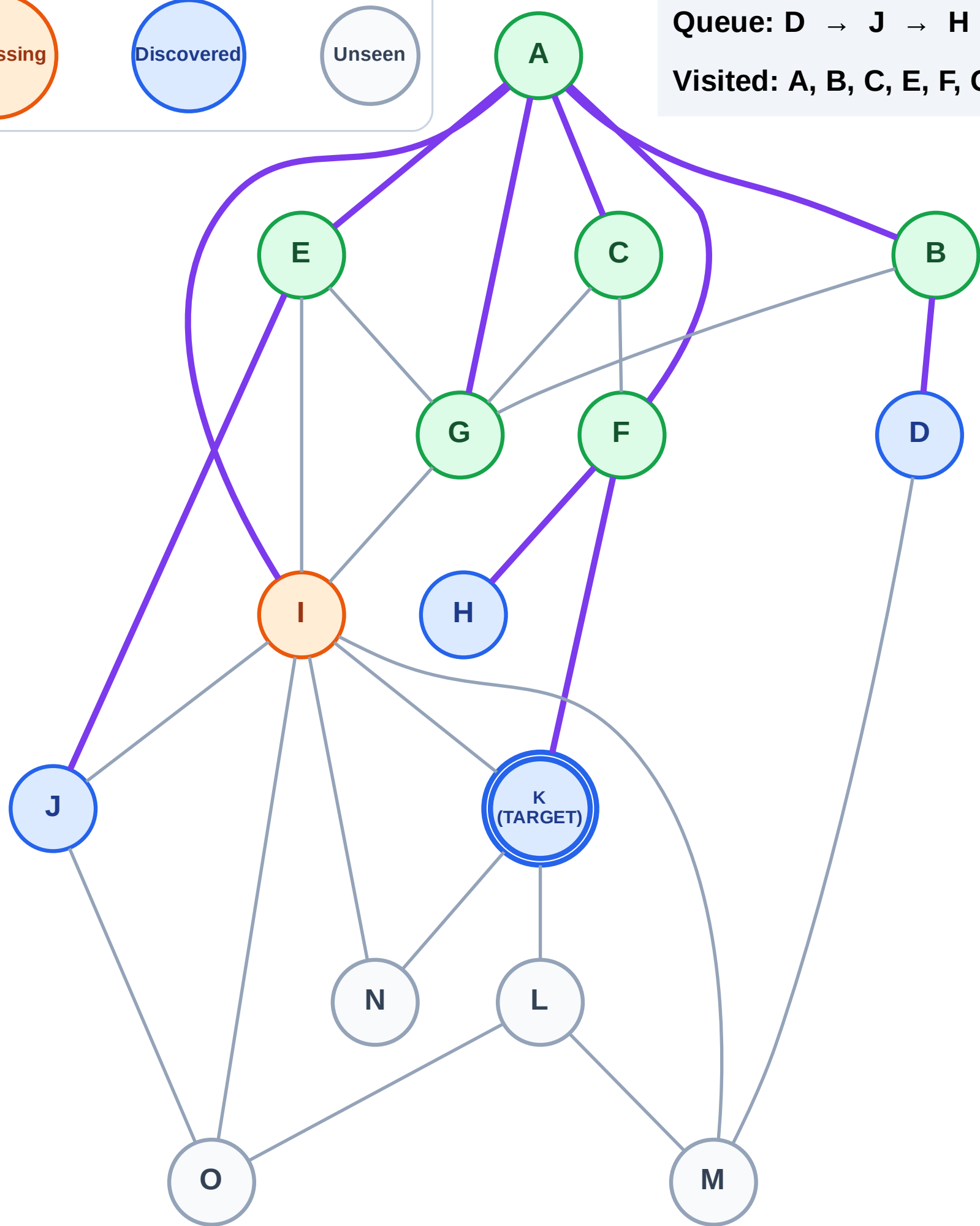
Processing

Discovered

Unseen

Queue: D → J → H → K

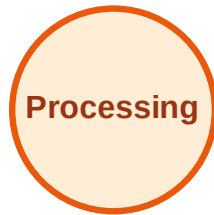
Visited: A, B, C, E, F, G



BFS Traversal

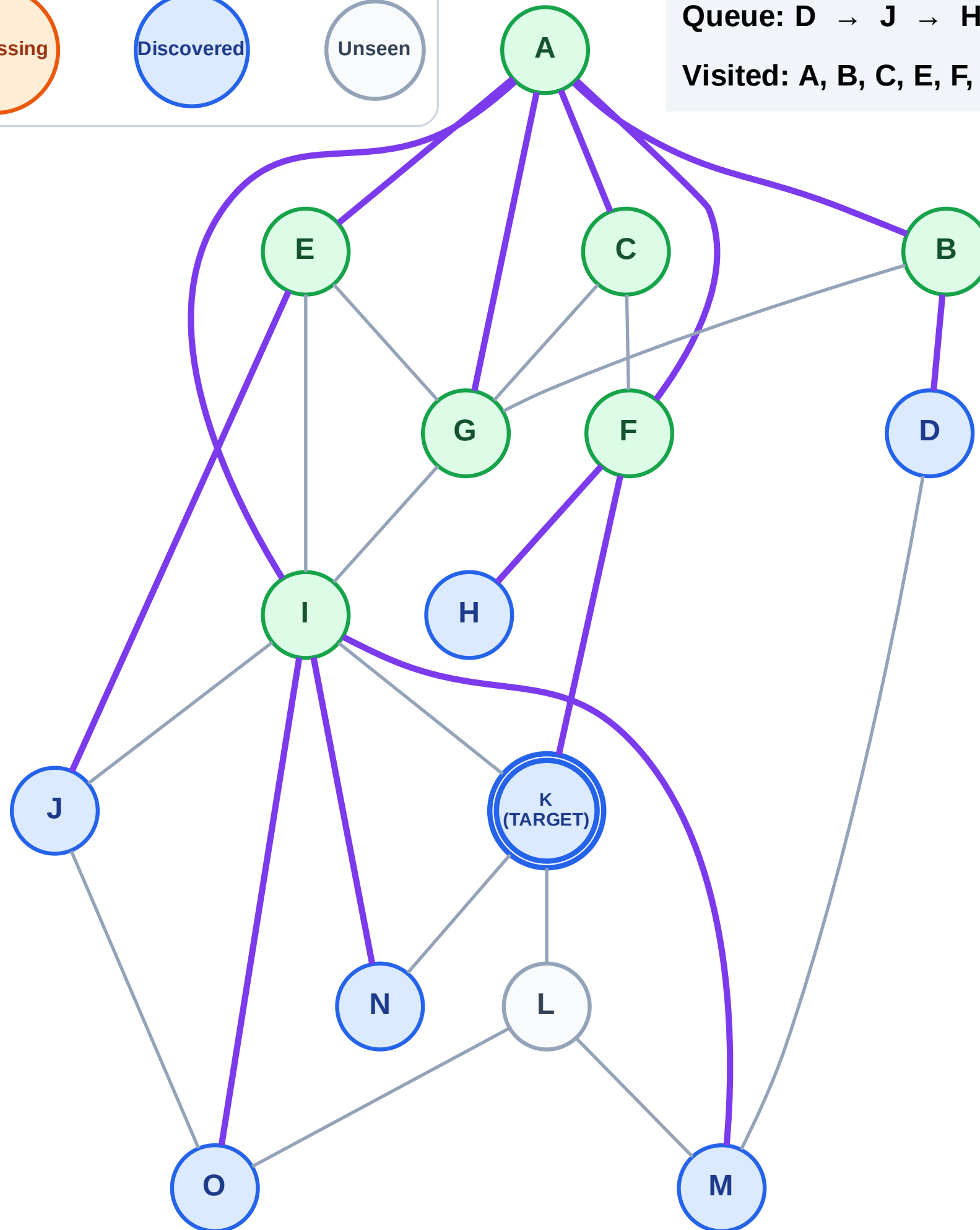
Step 15: Discover M, N, O from I

Legend



Queue: D → J → H → K → M → N → O

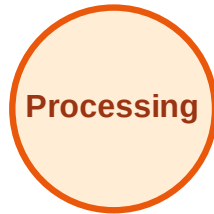
Visited: A, B, C, E, F, G, I



BFS Traversal

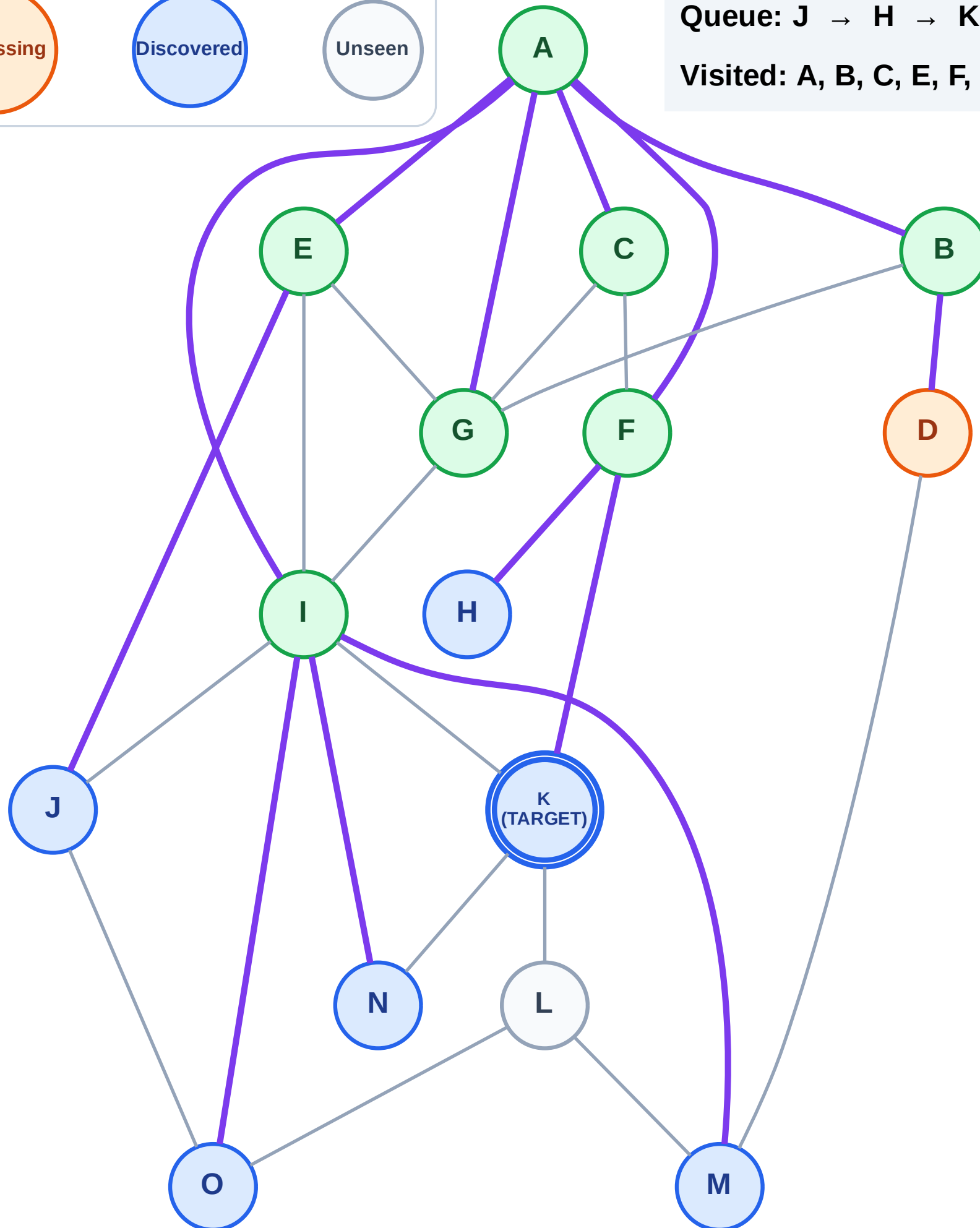
Step 16: Process node D

Legend



Queue: J → H → K → M → N → O

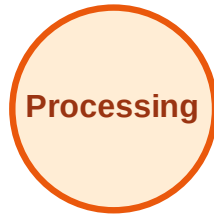
Visited: A, B, C, E, F, G, I



BFS Traversal

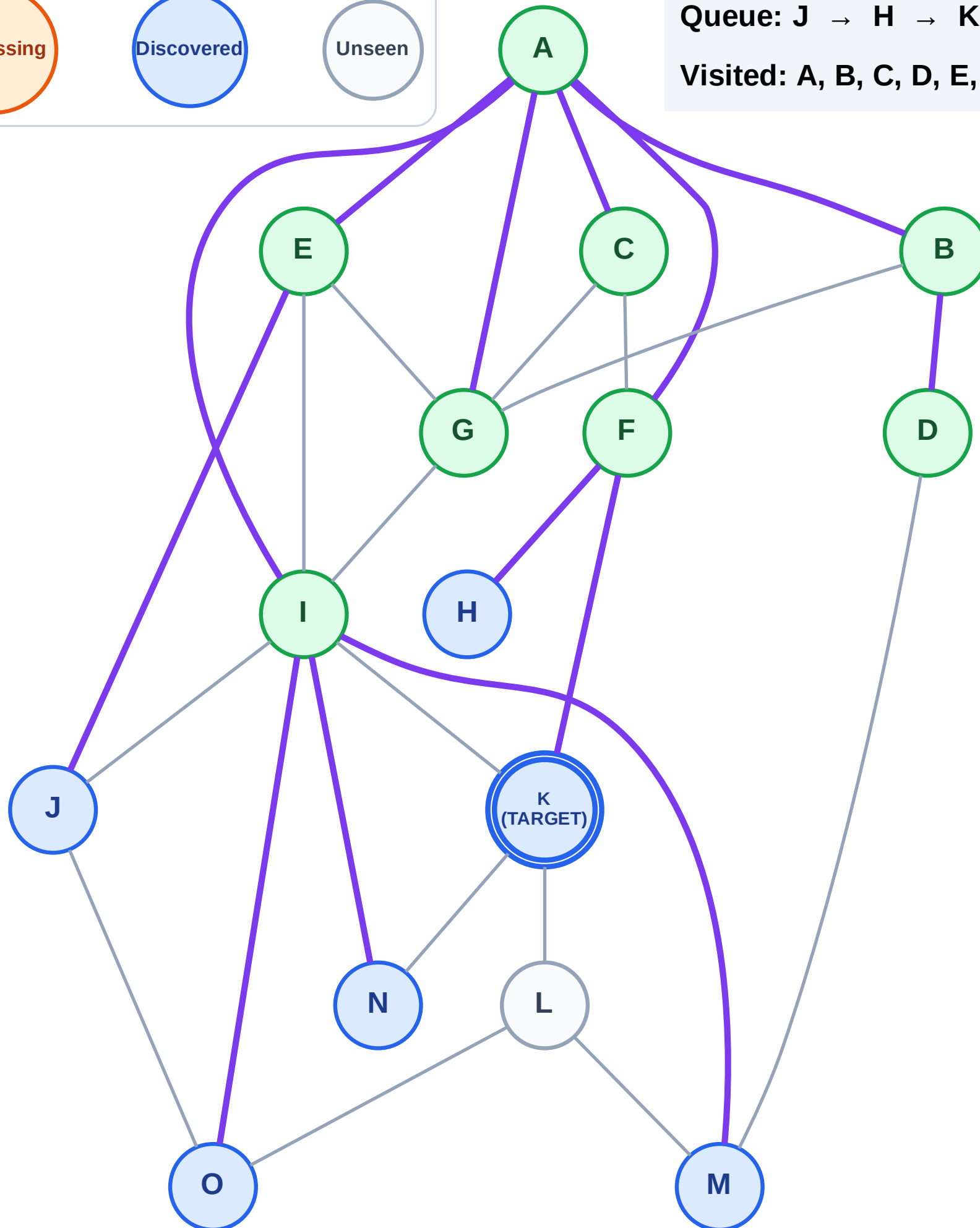
Step 17: No new neighbors from D

Legend



Queue: J → H → K → M → N → O

Visited: A, B, C, D, E, F, G, I



BFS Traversal

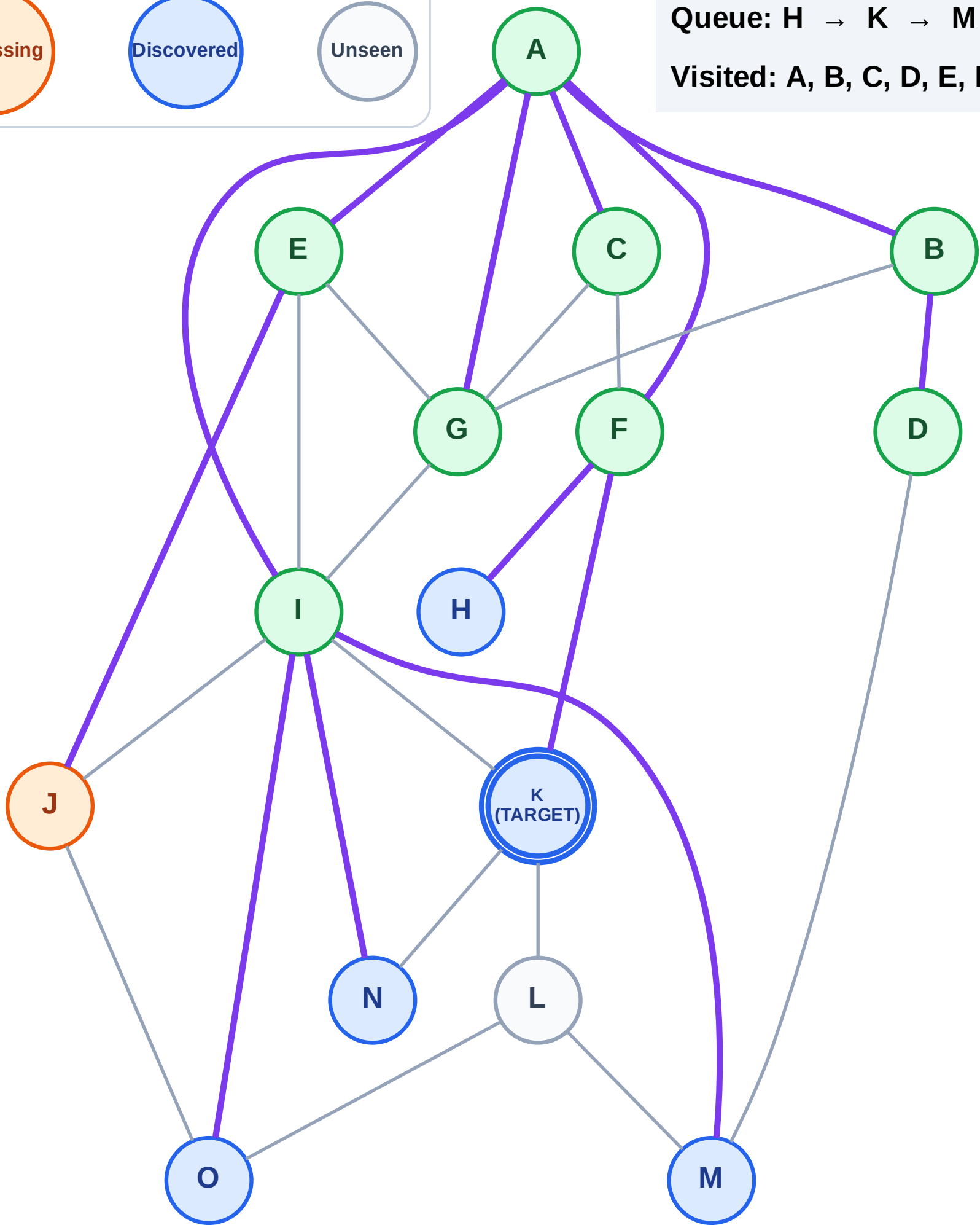
Step 18: Process node J

Legend



Queue: H → K → M → N → O

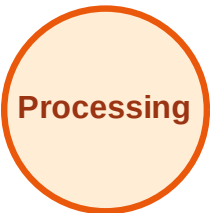
Visited: A, B, C, D, E, F, G, I



BFS Traversal

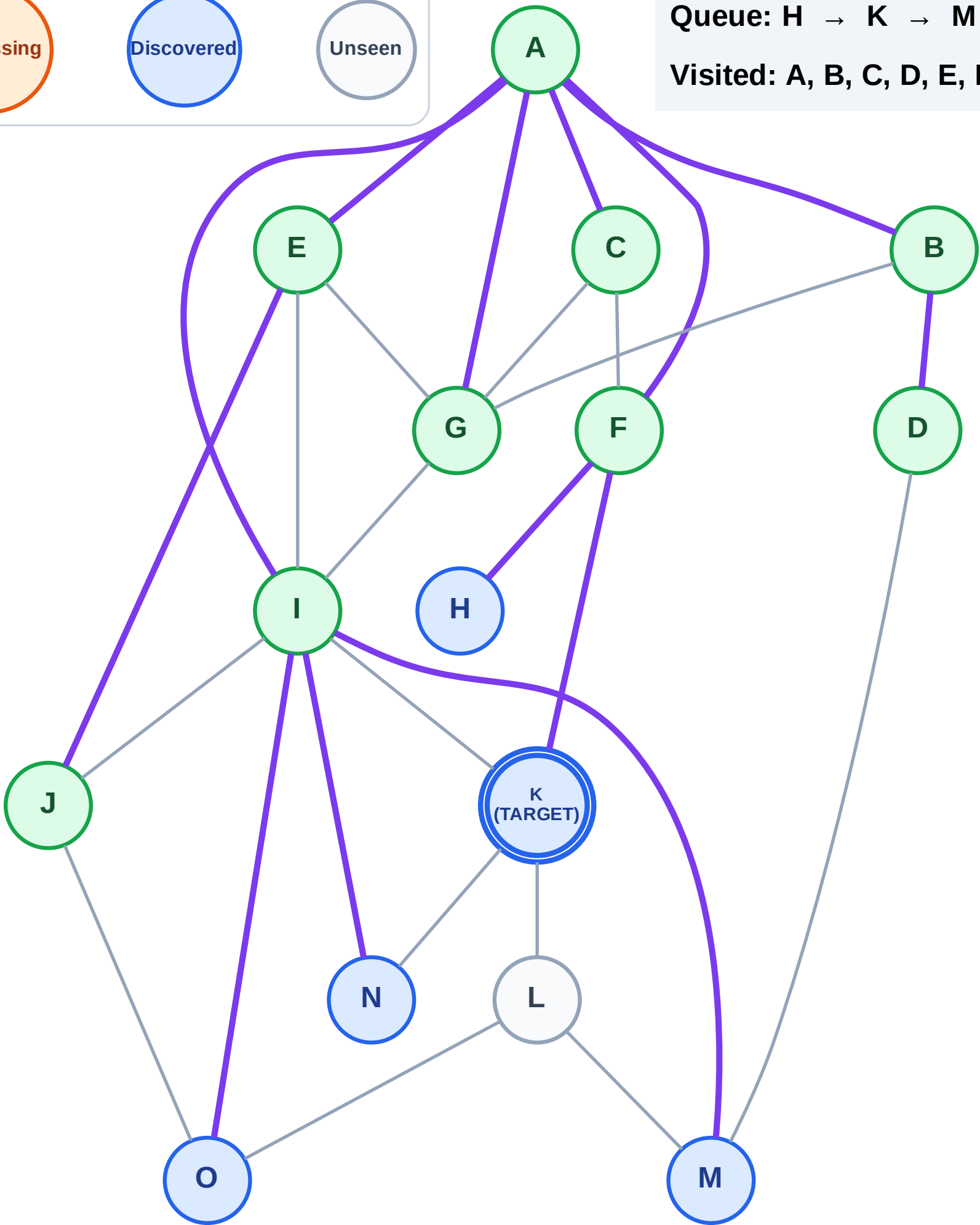
Step 19: No new neighbors from J

Legend



Queue: H → K → M → N → O

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

Step 20: Process node H

Legend

Complete

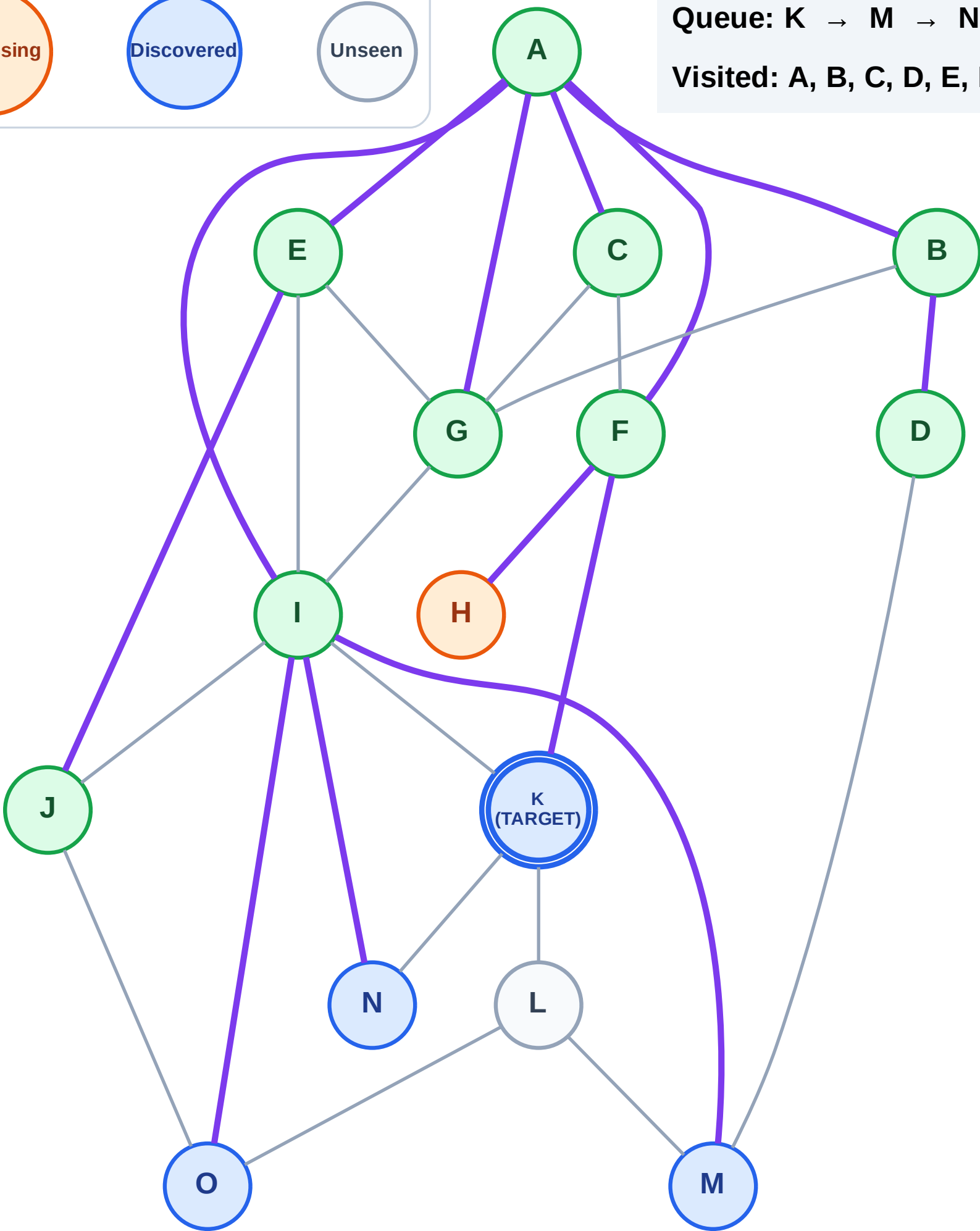
Processing

Discovered

Unseen

Queue: K → M → N → O

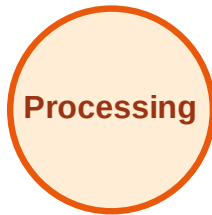
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

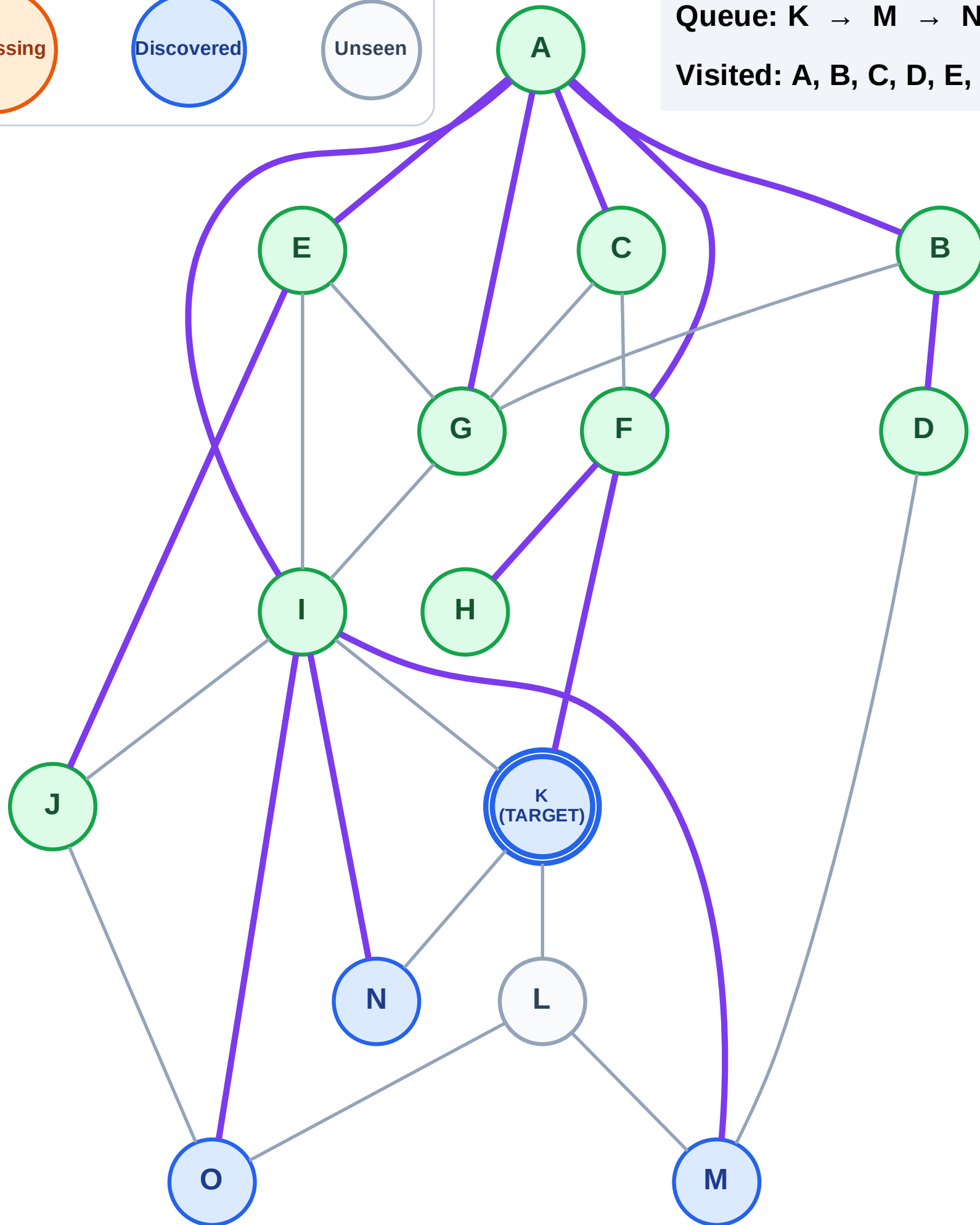
Step 21: No new neighbors from H

Legend



Queue: K → M → N → O

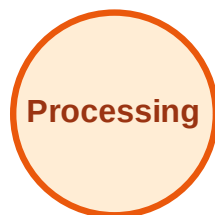
Visited: A, B, C, D, E, F, G, H, I, J



BFS Traversal

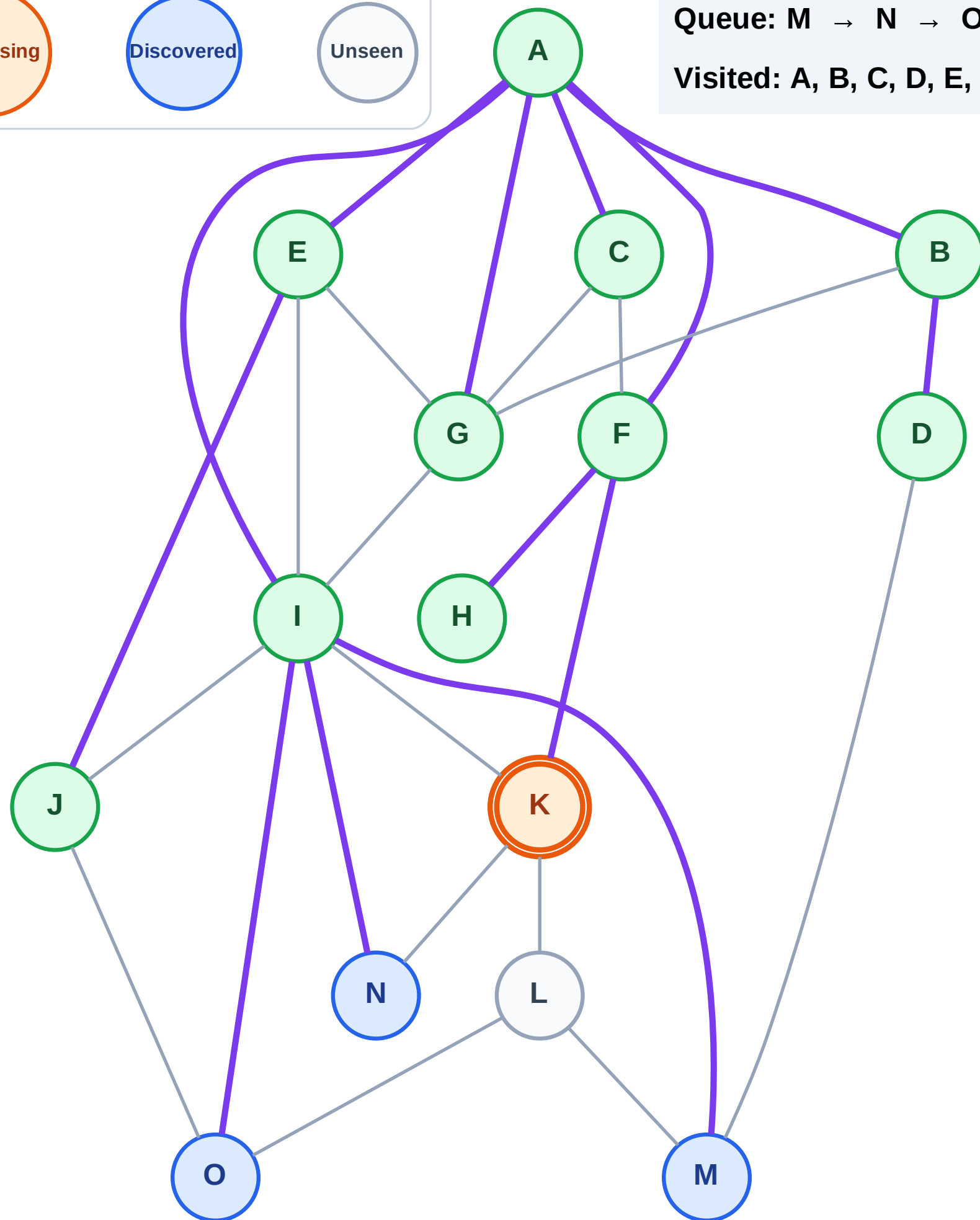
Step 22: Process node K

Legend



Queue: M → N → O

Visited: A, B, C, D, E, F, G, H, I, J



BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → F → K

Legend

Complete

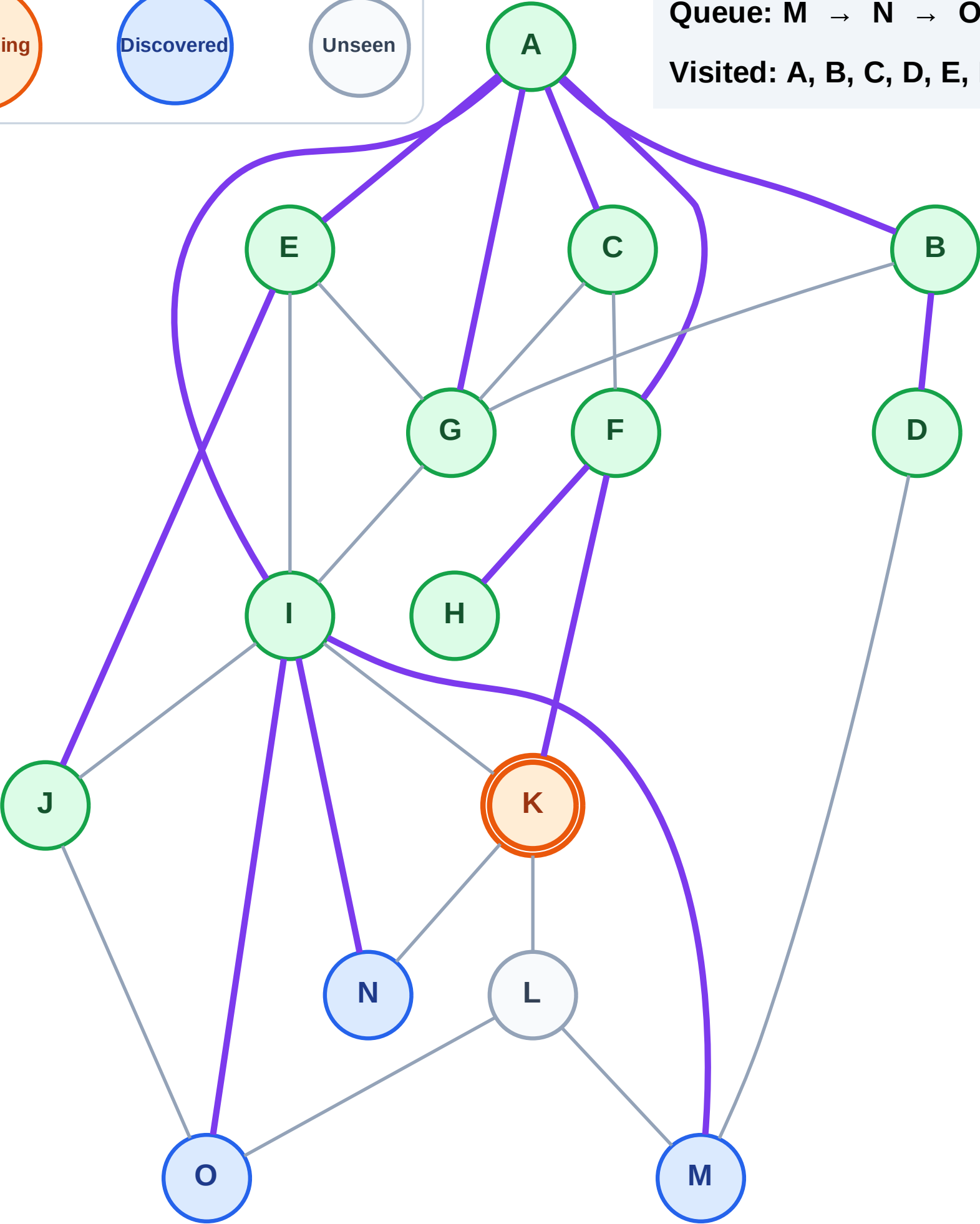
Processing

Discovered

Unseen

Queue: M → N → O

Visited: A, B, C, D, E, F, G, H, I, J, K



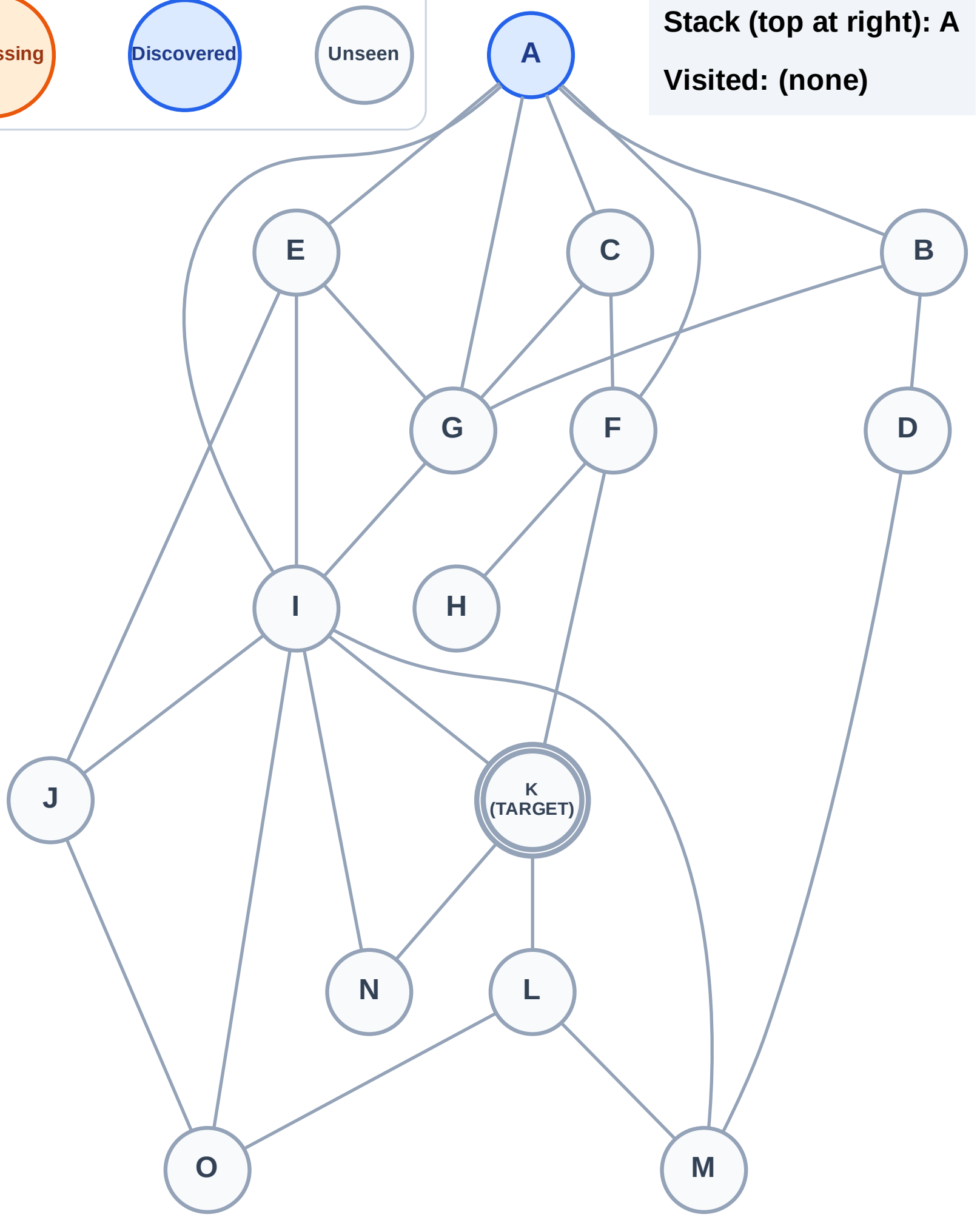
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

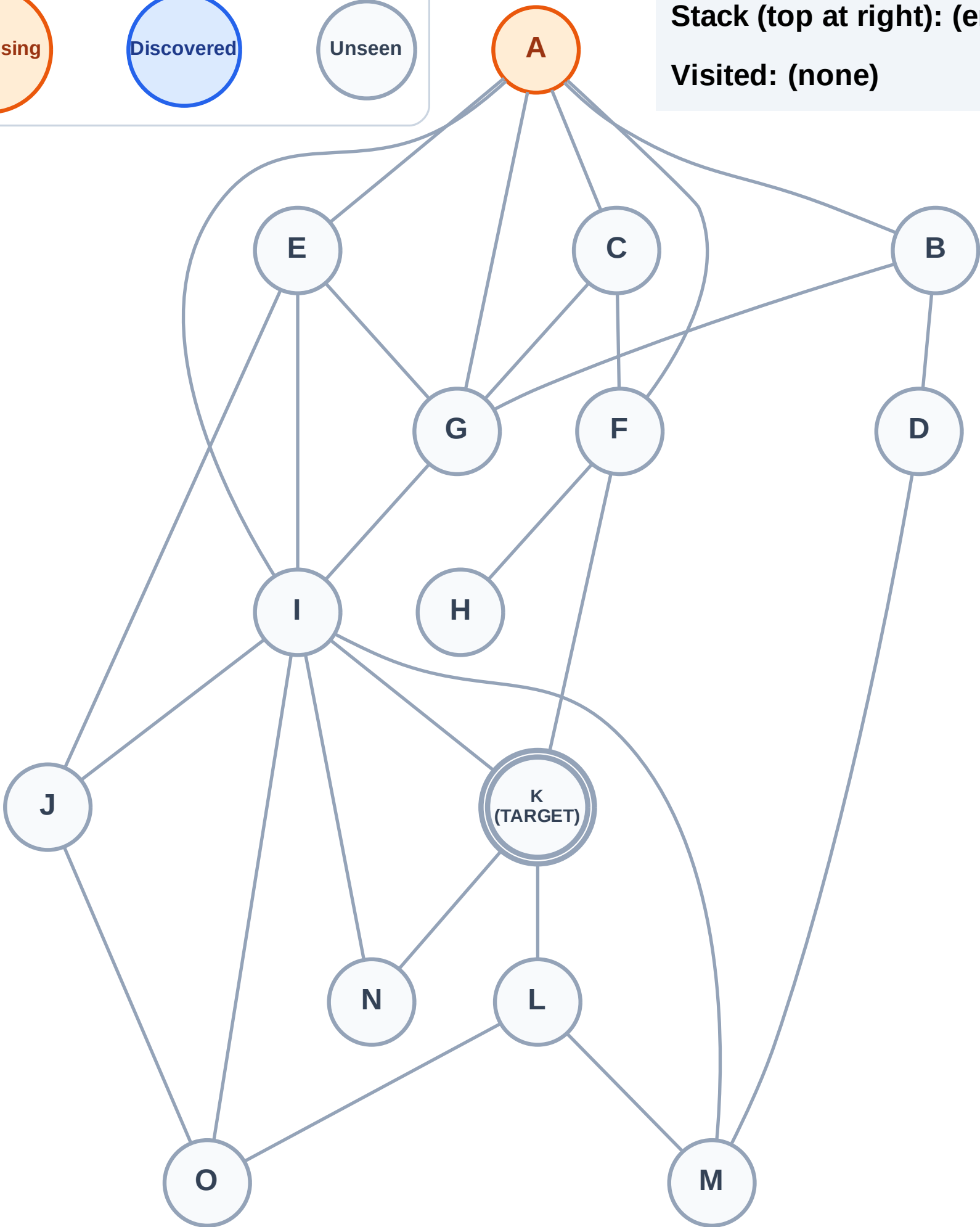
Complete

Processing

Discovered

Unseen

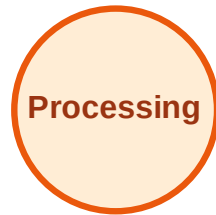
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

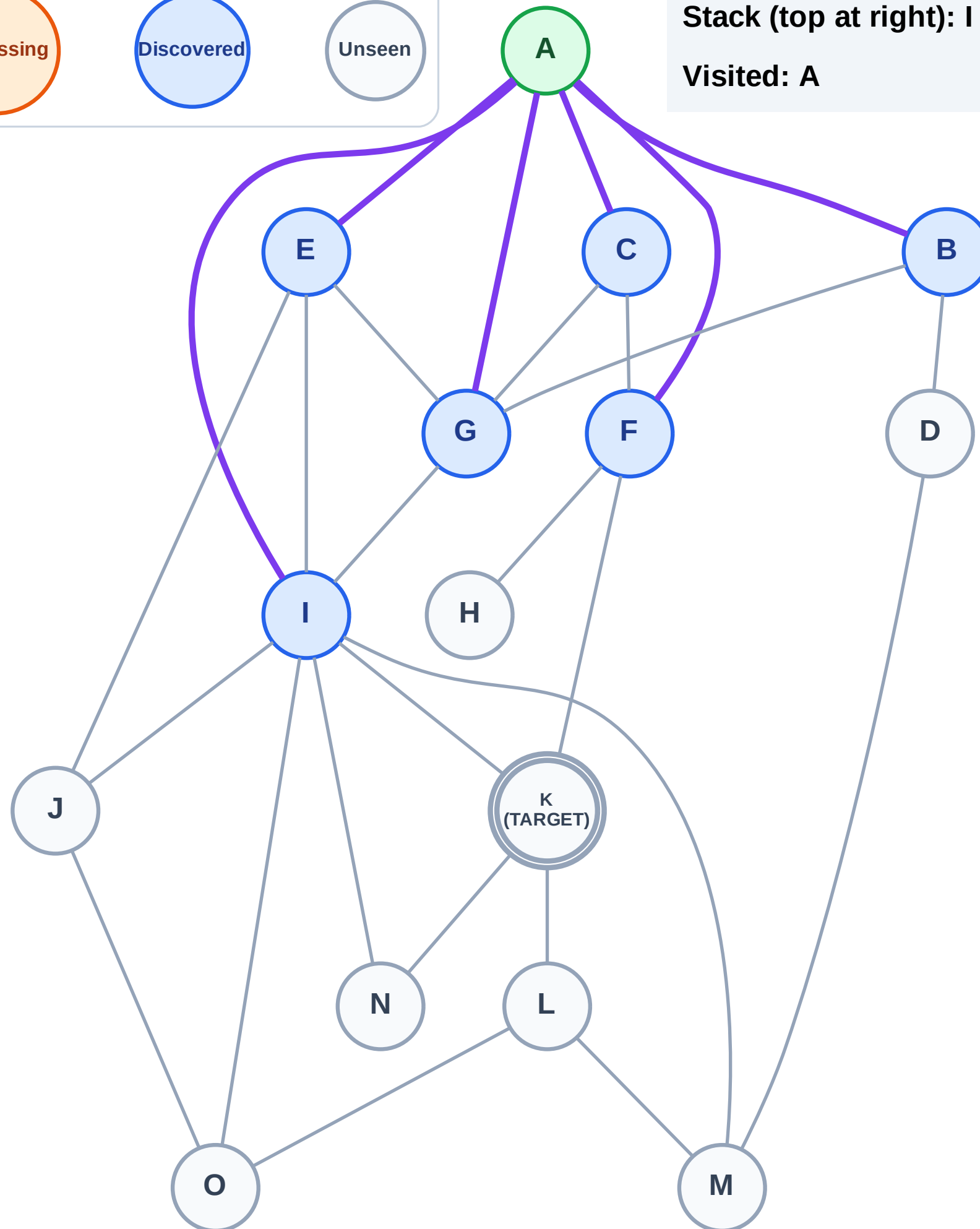
Step 3: Discover I, G, F, E, C, B from A

Legend

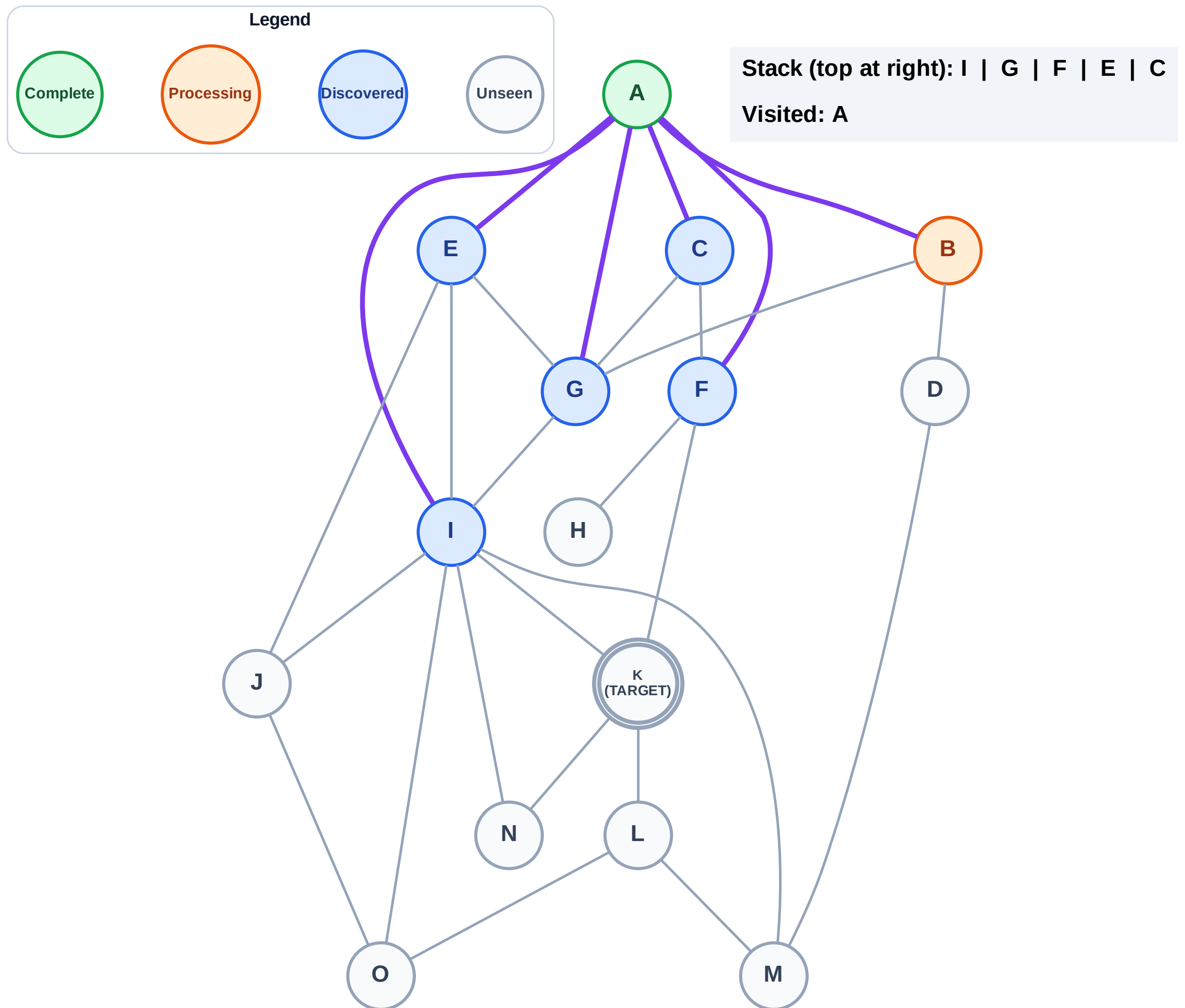


Stack (top at right): I | G | F | E | C | B

Visited: A



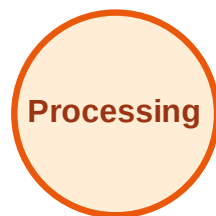
Step 4: Process node B



DFS Traversal

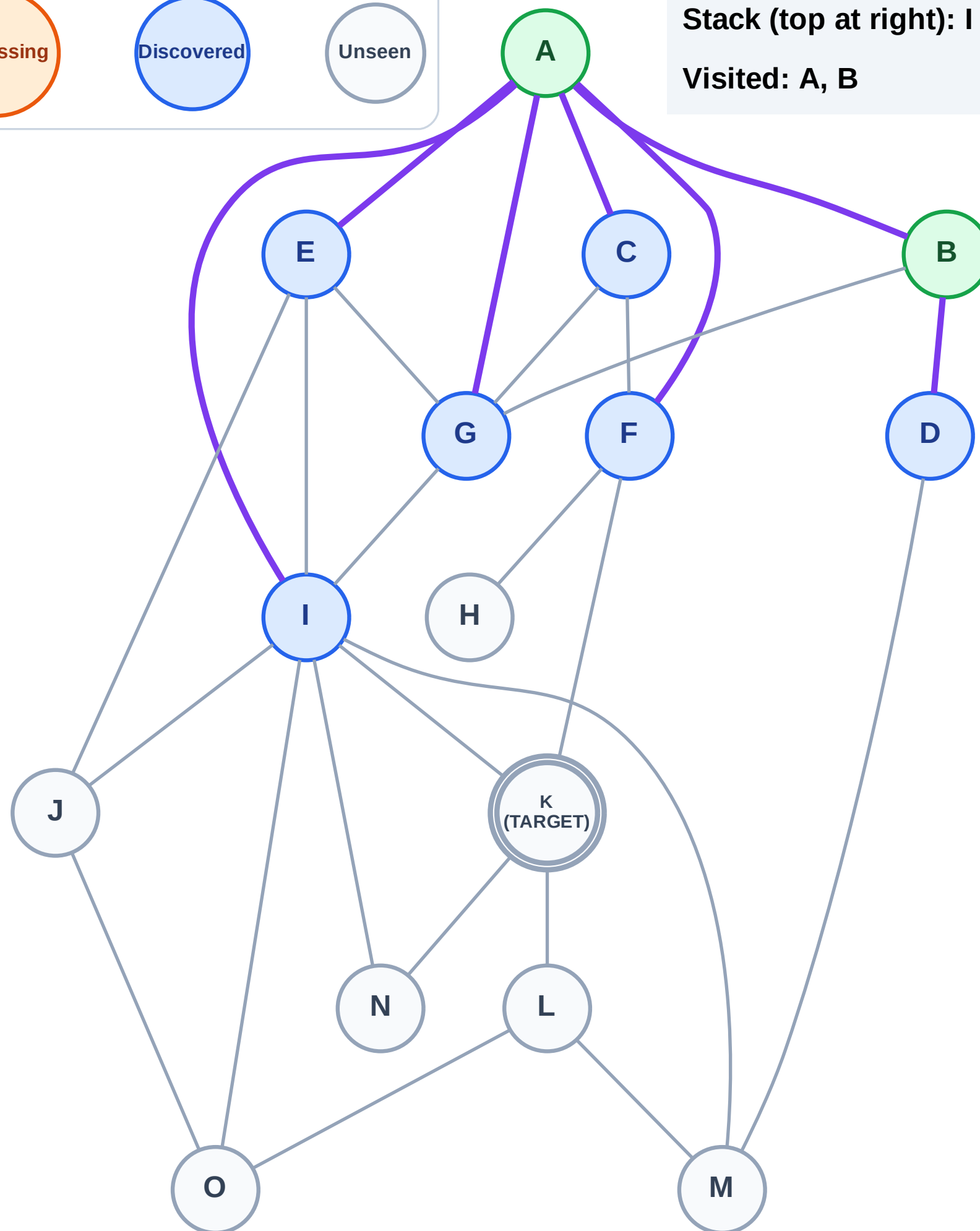
Step 5: Discover D from B

Legend



Stack (top at right): I | G | F | E | C | D

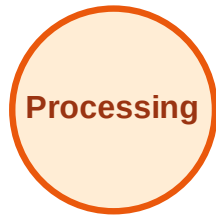
Visited: A, B



DFS Traversal

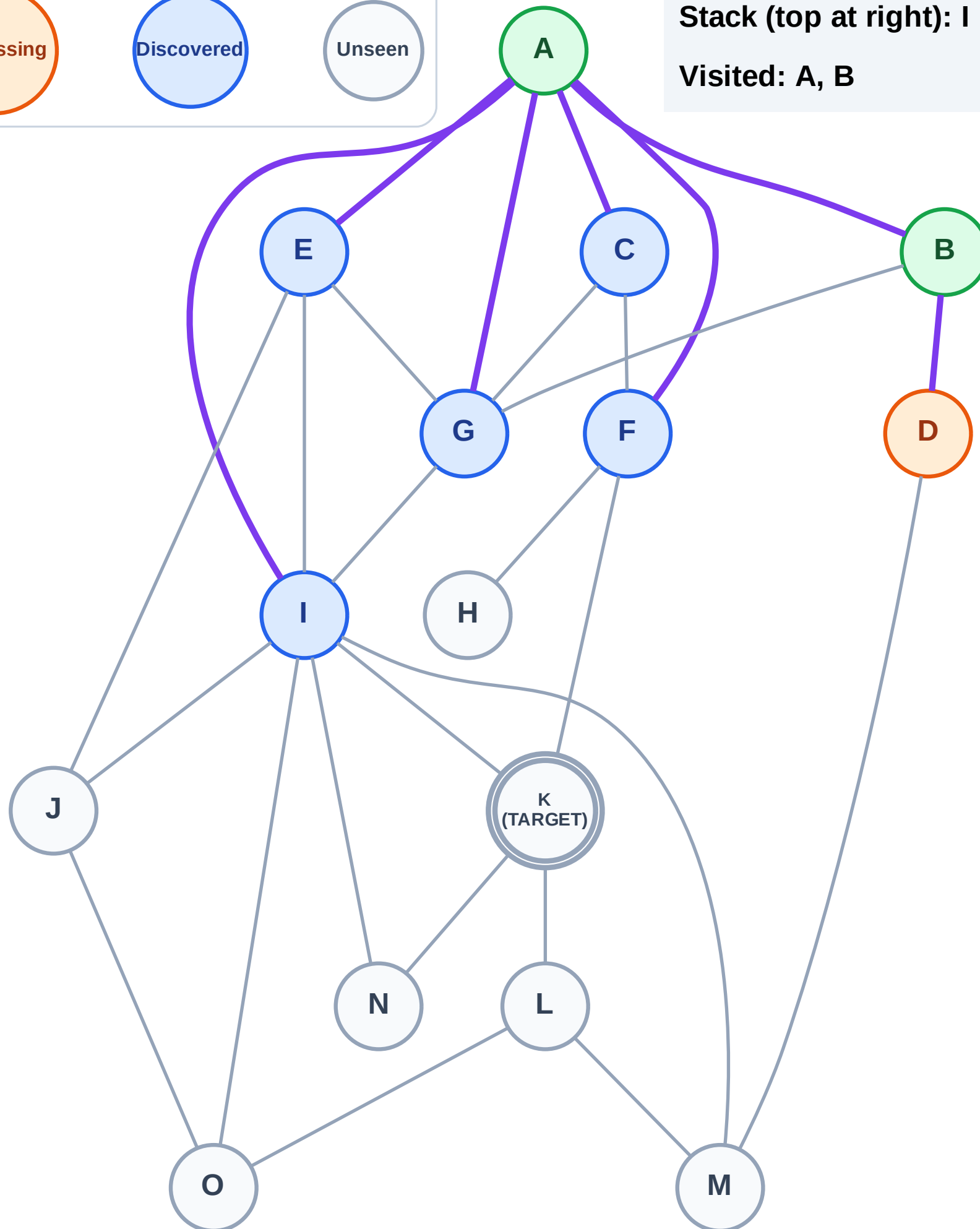
Step 6: Process node D

Legend



Stack (top at right): I | G | F | E | C

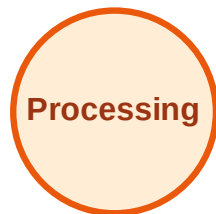
Visited: A, B



DFS Traversal

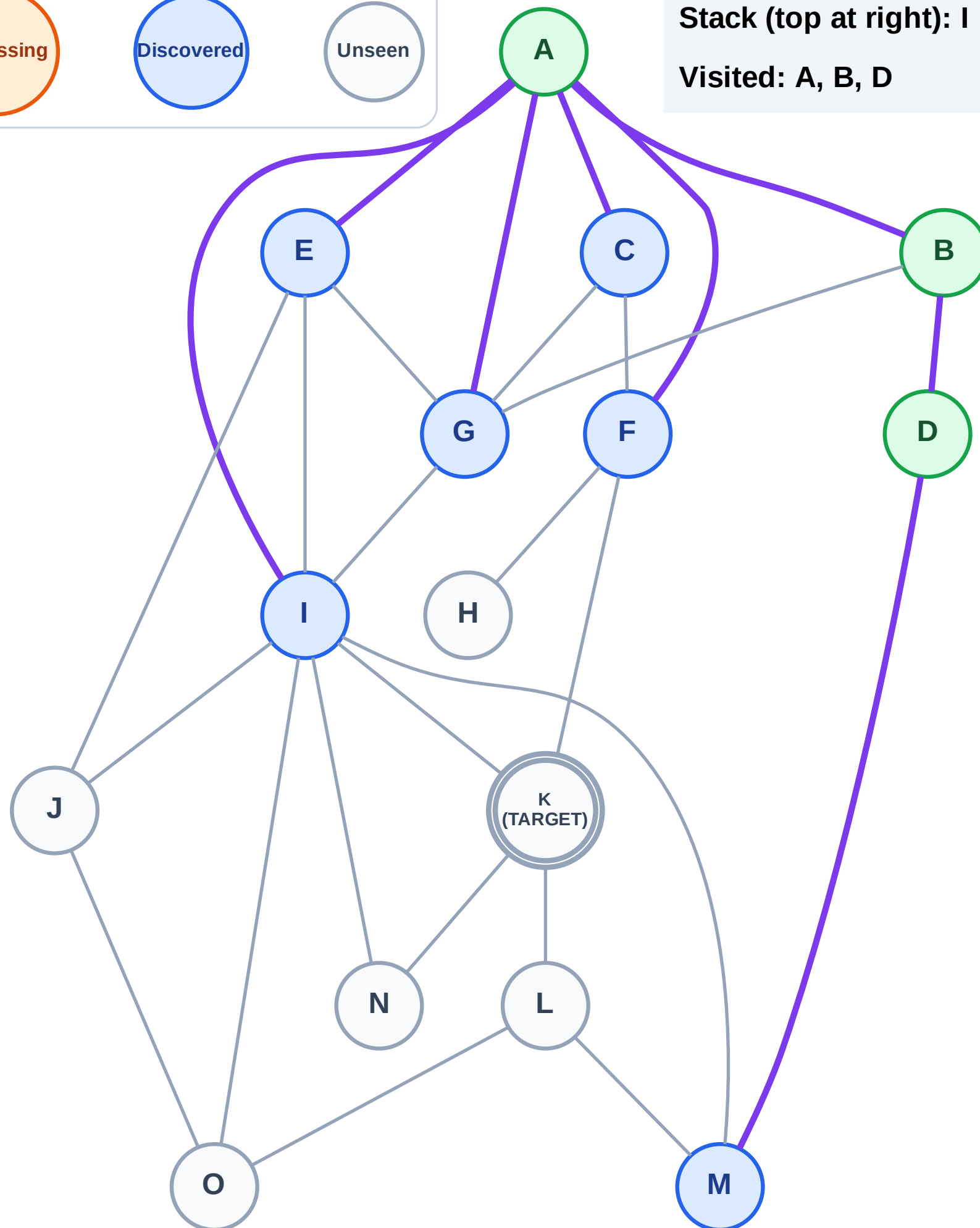
Step 7: Discover M from D

Legend



Stack (top at right): I | G | F | E | C | M

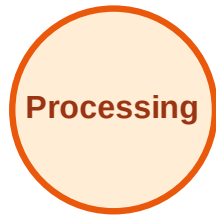
Visited: A, B, D



DFS Traversal

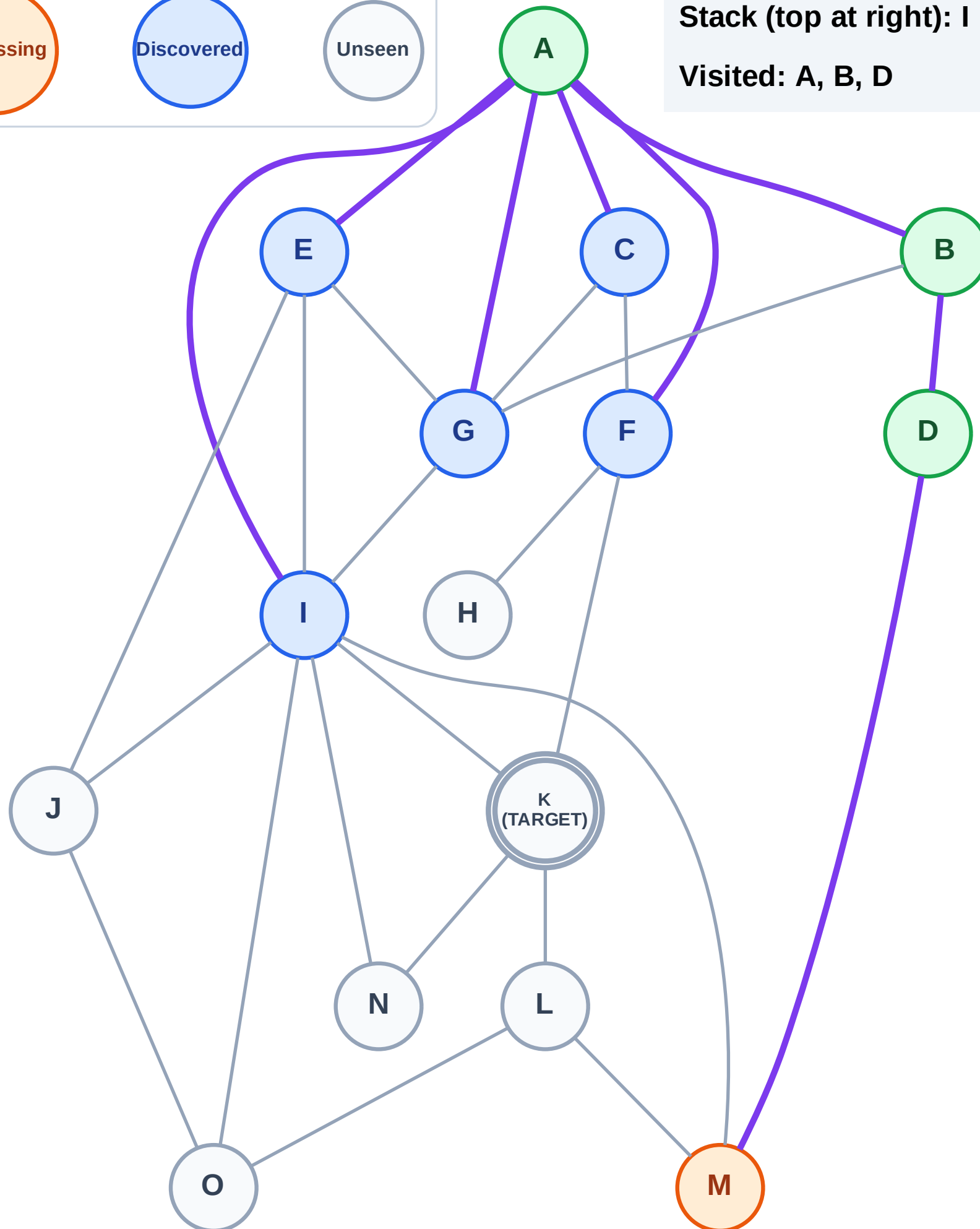
Step 8: Process node M

Legend



Stack (top at right): I | G | F | E | C

Visited: A, B, D



DFS Traversal

Step 9: Discover L from M

Legend

Complete

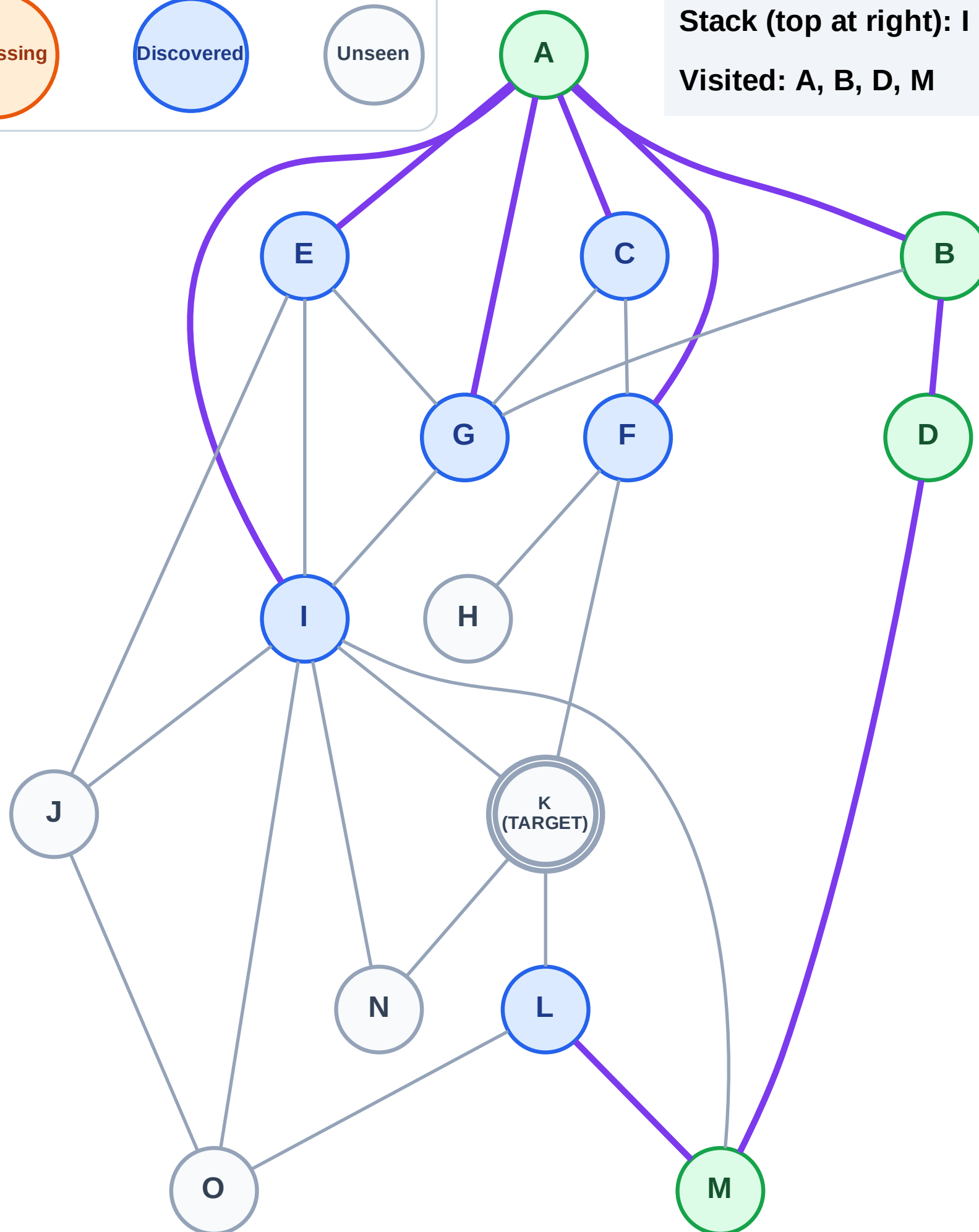
Processing

Discovered

Unseen

Stack (top at right): I | G | F | E | C | L

Visited: A, B, D, M



DFS Traversal

Step 10: Process node L

Legend

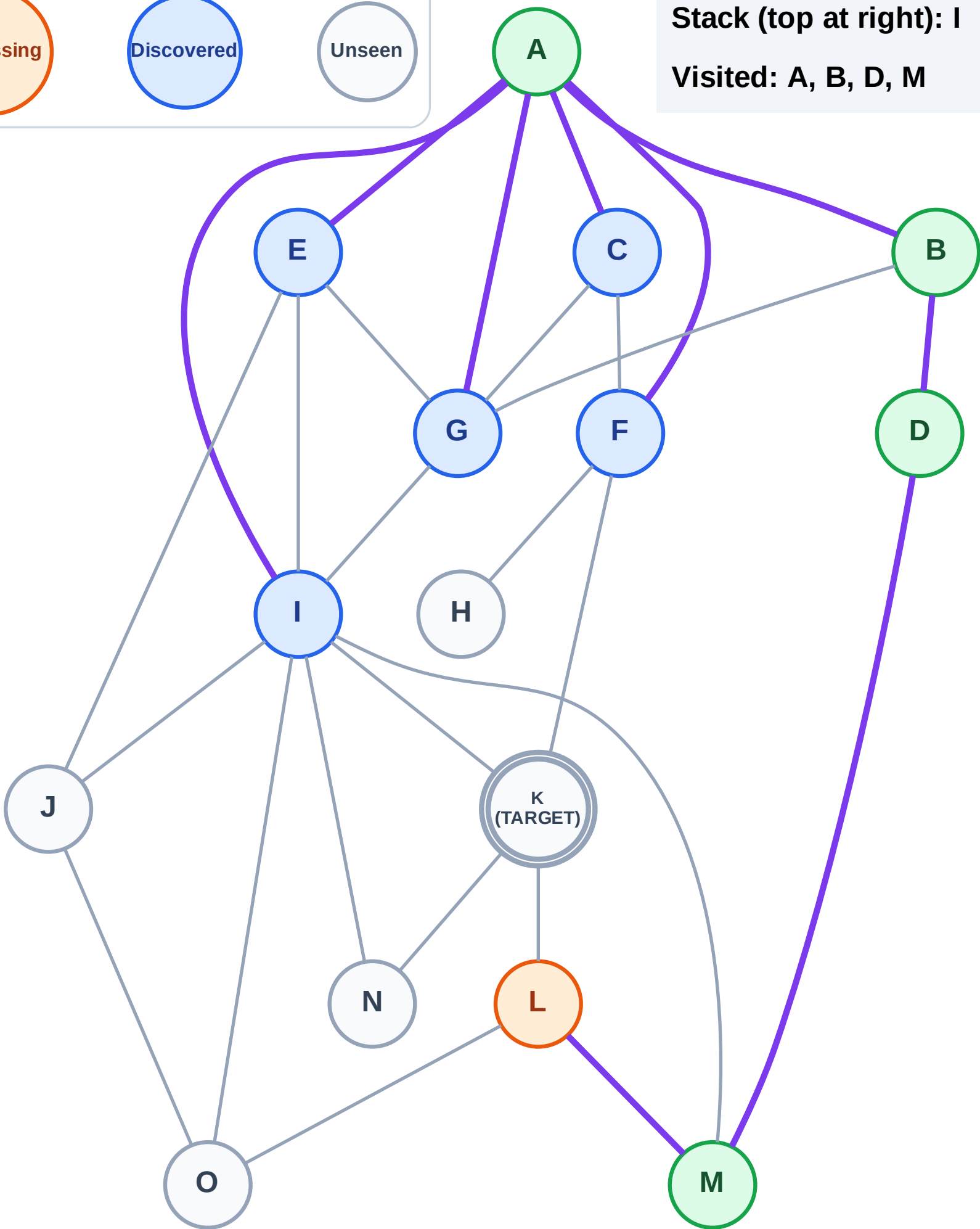
Complete

Processing

Discovered

Unseen

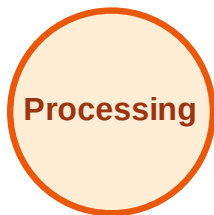
Stack (top at right): I | G | F | E | C
Visited: A, B, D, M



DFS Traversal

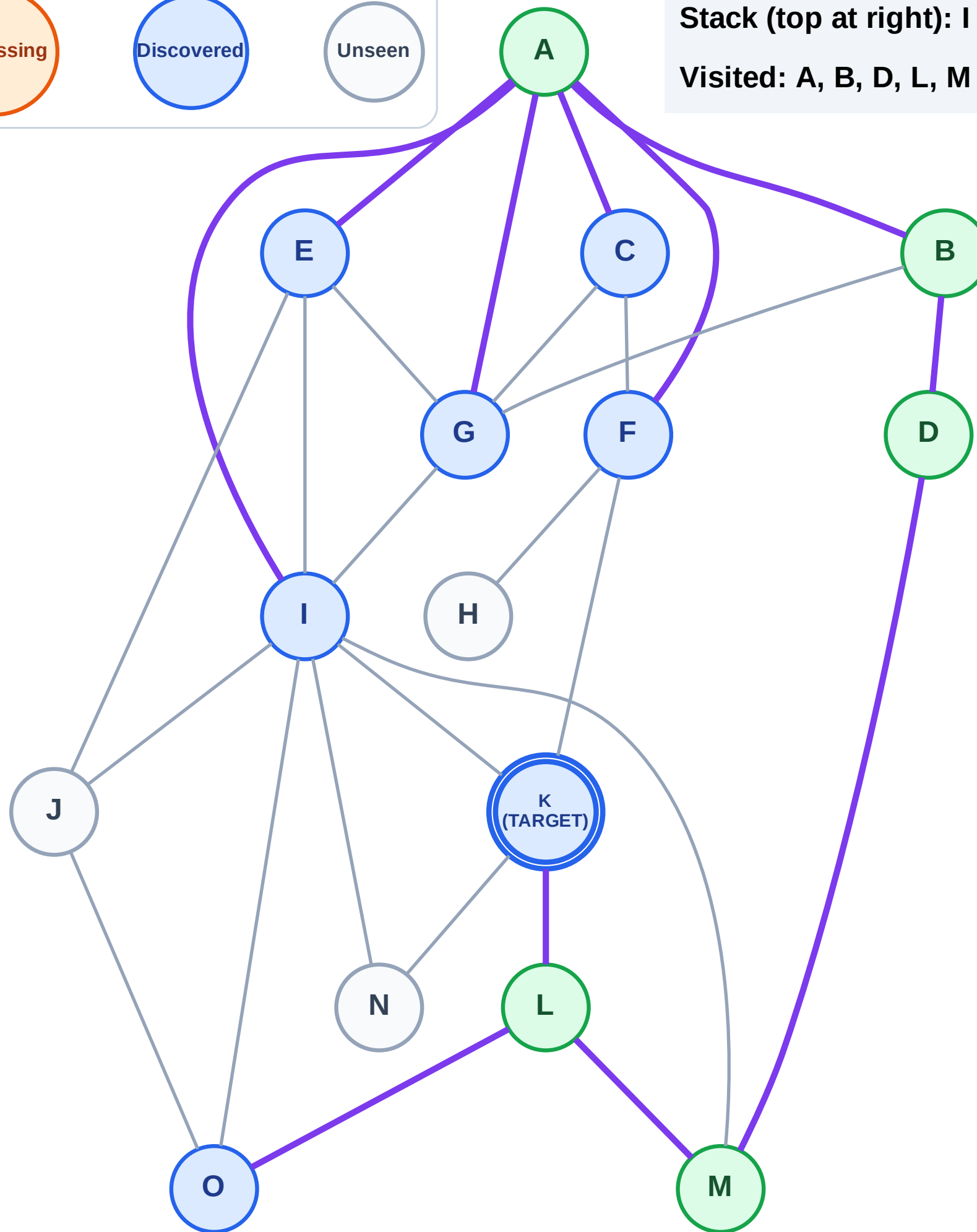
Step 11: Discover O, K from L

Legend



Stack (top at right): I | G | F | E | C | O | K

Visited: A, B, D, L, M



DFS Traversal

Step 12: Process node K

Legend

Complete

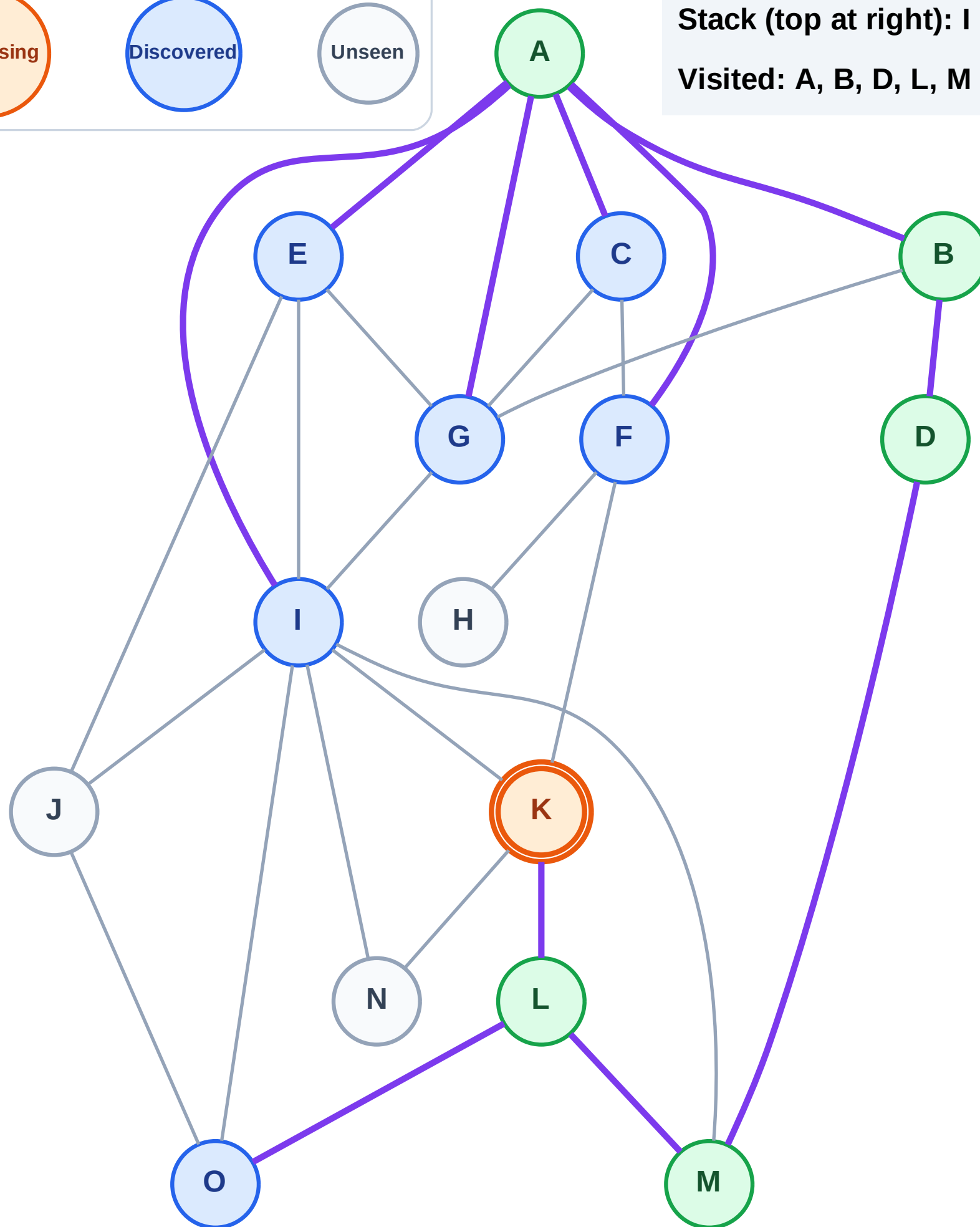
Processing

Discovered

Unseen

Stack (top at right): I | G | F | E | C | O

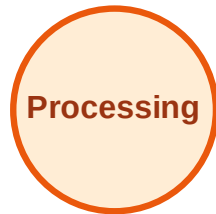
Visited: A, B, D, L, M



DFS Traversal

Step 13: TARGET NODE K REACHED! Path (5 hops): A → B → D → M → L → K

Legend



Stack (top at right): I | G | F | E | C | O

Visited: A, B, D, K, L, M

